Chapter 10: Analysis and Processing of Random Signals

10.1 Power Spectral Density

10.1

a) $S_X(f) = \mathcal{F}\left[g\left(\frac{\tau}{T}\right)\right] = AT\left(\frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}}\right)^2$

Table in Appendix B.

b) $S_X(f) = g\left(\frac{f}{W}\right)$

$R_X(\tau) = AW\left(\frac{\sin\frac{W\tau}{2}}{\frac{W\tau}{2}}\right)^2$

10.2

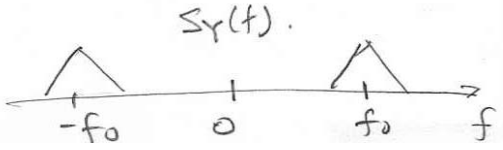
$\mathcal{F}\left[p\left(\frac{\tau}{T}\right)\right] = 2AT\frac{\sin 2\pi fT}{2\pi fT}$ which is negative for some values of f. Since power spectral densities are always non-negative, $p\left(\frac{\tau}{T}\right)$ is not a valid autocorrelation function.

10.3

$$\begin{aligned} S_Y(f) &= \mathcal{F}[R_X(\tau)\cos 2\pi f_0\tau] \\ &= \mathcal{F}\left[R_X(\tau)\left[\frac{e^{j2\pi f_0\tau} + e^{-j2\pi f_0\tau}}{2}\right]\right] \\ &= \frac{1}{2}\mathcal{F}[R_X(\tau)e^{j2\pi f_0\tau}] + \frac{1}{2}\mathcal{F}[R_X(\tau)e^{-j2\pi f_0\tau}] \\ &= \frac{1}{2}S_X(f - f_0) + \frac{1}{2}S_X(t + f_0) \end{aligned}$$

where

$S_X(f) = \mathcal{F}[R_X(\tau)]$



10.4

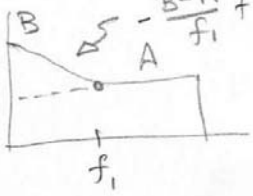
$S_X(f) = Ap\left(\frac{f}{f_2}\right) + (B-A)g\left(\frac{f}{f_1}\right)$ where $p(x)$ is as in Fig. P10.2 and $g(x)$ as in Fig. P10.1.
Then from Table C in Appendix B:

$$R_X(\tau) = 2Af_2\frac{\sin 2\pi f_2\tau}{2\pi f_2\tau} + (B-A)f_1\left(\frac{\sin 2\pi f_1\tau/2}{2\pi f_1\tau/2}\right)^2$$

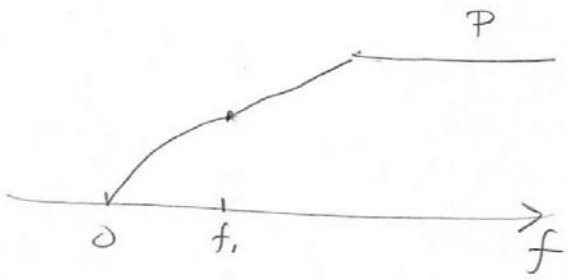
— continued —

(b) $$P = \int_{-\infty}^{\infty} S_X(f)\,df = A(2f_2) + (B-A)(2f_1)\tfrac{1}{2}$$

$$= 2Af_2 + (B-A)f_1$$



(c) $$2\int_0^{f_0} S_X(f)\,df = \begin{cases} -2\left(\frac{B-A}{f_1}\right)\frac{f_0^2}{2} + Bf_0 & 0 < f_0 < f_1 \\ 2\left[\frac{B-A}{2}f_1 + A(f_0 - f_1)\right] & f_1 < f_0 < f_2 \end{cases}$$



10.5

$$R_X(\tau) = \delta_X^2 e^{-2\alpha^2\tau^2},$$

a) From Fourier transform table we have: $e^{-\frac{t^2}{2\beta^2}} \xleftrightarrow{F} \beta\sqrt{2\pi}\, e^{-\frac{\beta^2(2\pi f)^2}{2}}$

in our case $\beta^2 = \frac{1}{4\alpha^2}$ & $\beta = \frac{1}{2\alpha}$

Therefore: $S_X(f) = \delta_X^2 \times \frac{\sqrt{2\pi}}{2\alpha} e^{\frac{-f^2(2\pi)^2}{2(4\alpha^2)}} = \delta_X^2 \frac{1}{\frac{\alpha}{\pi}\sqrt{2\pi}} e^{-\frac{f^2}{2(\frac{\alpha}{\pi})^2}}$

as it can be seen $S_X(f)$ is δ_X^2 times of a Gaussian function with mean zero and variance $\frac{\alpha^2}{\pi^2}$.

b) $$\int_{-\infty}^{-K\alpha} S_X(f)df + \int_{K\alpha}^{+\infty} S_X(f)df = 2\times\int_{K\alpha}^{\infty} S_X(f)df = 2\delta_X^2 Q\left(\frac{K\alpha}{\frac{\alpha}{\pi}}\right) = 2\delta_X^2 Q(K\pi)$$

in which Q is the Q-function.

Therefor: $K=1$, $2\delta_X^2 Q(\pi) = 2\delta_X^2 \times 8.4\times10^{-4}$

$K=2$, $2\delta_X^2 (Q(2\pi)) = 2\delta_X^2 \times 1.65\times10^{-10}$

$K=3$, $2\delta_X^2 Q(3\pi) \approx 2\delta_X^2 \times Q(9.5) = 2\delta_X^2 \times 1.05\times10^{-21}$

10.6

~~7.5~~ From Example 10.4

$$\begin{aligned} R_Z(\tau) &= R_X(\tau) + R_{YX}(\tau) + R_{XY}(\tau) + R_Y(\tau) \\ \Rightarrow S_Z(f) &= S_X(f) + S_{YX}(f) + S_{XY}(f) + S_Y(f) \end{aligned}$$

If $X(t)$ and $Y(t)$ are orthogonal processes, then

$$R_{XY}(\tau) = R_{XY}(\tau) = 0 \Rightarrow S_Z(f) = S_X(f) + S_Y(f)$$

10.7

~~7.6~~ a)

$$\begin{aligned} R_{XY}(\tau) &= \mathcal{E}[X(t+\tau)Y(t)] \\ &= \mathcal{E}[Y(t)X(t+\tau)] \\ &= R_{YX}(-\tau) \end{aligned}$$

b)

$$\begin{aligned} S_{XY}(t) &= \mathcal{F}[R_{XY}(\tau)] = \int_{-\infty}^{\infty} R_{XY}(\tau)e^{-j2\pi f\tau}d\tau \\ &= \int_{-\infty}^{\infty} R_{YX}(-\tau)e^{-j2\pi f\tau}d\tau \\ &= \int_{-\infty}^{\infty} R_{YX}(\tau')e^{+j2\pi f\tau'}d\tau' \\ &= S_{YX}^*(f) \end{aligned}$$

10.8

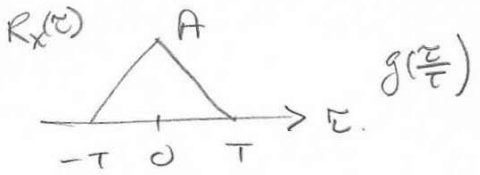
~~7.7~~ a)

$$\begin{aligned} R_{XY}(\tau) &= \mathcal{E}[X(t+\tau)[X(t) - X(t-d)]] \\ &= R_X(\tau) - R_X(\tau+d) \\ S_{XY}(f) &= \mathcal{F}[R_X(\tau)] - \mathcal{F}[R_X(\tau+d)] = S_X(f) - S_X(f)e^{j2\pi fd} \end{aligned}$$

b)

$$\begin{aligned} R_Y(\tau) &= \mathcal{E}[(X(t+\tau) - X(t+\tau-d))(X(t) - X(t-d))] \\ &= R_X(\tau) - R_X(\tau+d) - R_X(\tau-d) + R_X(\tau) \\ &= 2R_X(\tau) - R_X(\tau+d) - R_X(\tau-d) \\ S_Y(f) &= 2S_X(f) - S_X(f)e^{j2\pi fd} - S_X(f)e^{-j2\pi fd} \\ &= 2S_X(f)[1 - \cos 2\pi fd] \end{aligned}$$

10.9



(a)

$$R_{XY}(\tau) = g\left(\frac{\tau}{T}\right) - g\left(\frac{\tau+d}{T}\right)$$

$$S_{XY}(f) = S_X(f) - S_X(f)e^{j2\pi fd}$$

$$= AT\left(\frac{\sin \pi fT}{\pi fT}\right)^2 (1 - e^{-j2\pi fd})$$

(b)

$$R_Y(\tau) = 2g\left(\frac{\tau}{T}\right) - g\left(\frac{\tau+d}{T}\right) - g\left(\frac{\tau-d}{T}\right)$$

$$S_Y(f) = 2AT\left(\frac{\sin \pi fT}{\pi fT}\right)^2 (1 - \cos 2\pi fd)$$

10.10

$$\begin{aligned}
\mathcal{E}[Z(t)] &= \mathcal{E}[X(t)]\mathcal{E}[Y(t)] = m_X m_Y \\
R_Z(t, t+\tau) &= \mathcal{E}[X(t)Y(t)X(t+\tau)Y(t+\tau)] \\
&= \mathcal{E}[X(t)X(t+\tau)]\mathcal{E}[Y(t)Y(t+\tau)] \\
&= R_X(\tau)R_Y(\tau) \\
S_Z(f) &= \mathcal{F}[R_X(\tau)R_Y(\tau)] = S_X(f) \star S_Y(f)
\end{aligned}$$

10.11

$$X(t) = a\cos(2\pi f_0 t + \Theta), \quad \Theta \sim U(0, 2\pi]$$

from P.10.10 we have: $S_Z(f) = S_X(f) * S_Y(f)$

$$R_Z(\tau) = R_X(\tau) \cdot R_Y(\tau)$$

$$R_X(\tau) = E[X(t)X(t+\tau)] = \frac{a^2}{2\pi}\int_0^{2\pi} \cos(2\pi f_0 t + \Theta)\cos(2\pi f_0 t + 2\pi f\tau + \Theta)\,d\Theta$$

$$= \frac{a^2}{2\pi} \times \frac{1}{2}\left[\int_0^{2\pi} \cos(2\pi f_0\tau)\,d\Theta + \int_0^{2\pi} \cos(4\pi f_0 t + 2\pi f_0\tau + 2\Theta)\,d\Theta\right]$$

$$= \frac{a^2}{2\pi} \times \frac{1}{2} \times 2\pi \times \cos(2\pi f_0\tau) + 0 = \frac{a^2}{2}\cos(2\pi f_0\tau)$$

$$S_X(f) = \mathcal{F}\{R_X(\tau)\} = \frac{1}{2}\{\delta(f - f_0) + \delta(f + f_0)\}$$

Therefore

$$S_Z(f) = \frac{1}{2}[S_Y(f - f_0) + S_Y(f + f_0)]$$

and

$$R_Z(\tau) = \frac{a^2}{2}\cos(2\pi f_0\tau)R_Y(\tau)$$

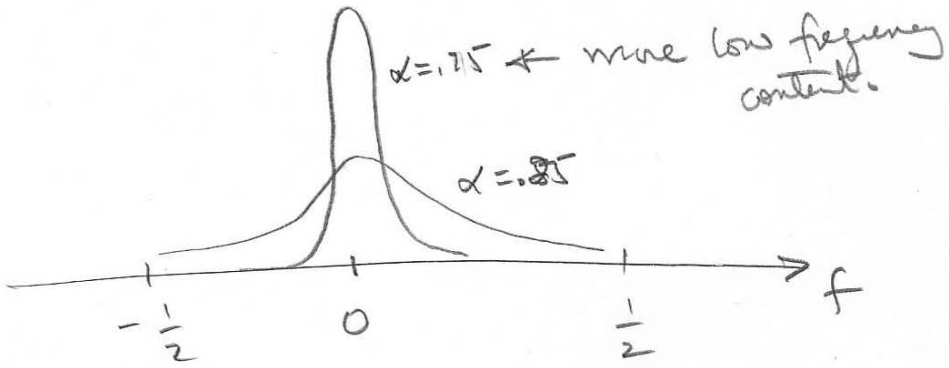
10.12 $R_X(k) = 4\alpha^{|k|} \quad |\alpha| < 1.$

(a) $$S_X(f) = 4\sum_{k=-\infty}^{\infty} \alpha^{|k|} e^{-j2\pi fk}$$

$$= 4 + 4\sum_{k=1}^{\infty} \alpha^k e^{-j2\pi fk} + 4\sum_{k=-\infty}^{-1} \left(\frac{1}{\alpha}\right)^k e^{-j2\pi fk}$$

$$= 4 + \frac{\alpha e^{-j2\pi f}}{1-\alpha e^{-j2\pi f}} + \frac{4\alpha e^{j2\pi f}}{1-\alpha e^{j2\pi f}}$$

$$= \frac{4(1-\alpha^2)}{1+\alpha^2-2\alpha\cos 2\pi f}$$



10.13

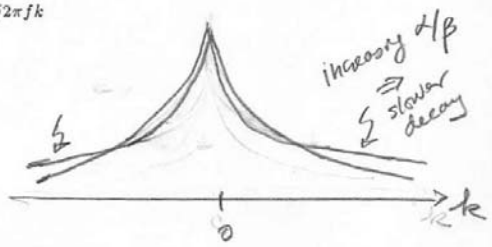
$$\sum_{k=-\infty}^{\infty} \alpha^{|k|} e^{-j2\pi fk} = 1 + \sum_{k=1}^{\infty} \alpha^k e^{-j2\pi fk} + \sum_{k=-\infty}^{-1} \left(\frac{1}{\alpha}\right)^k e^{-j2\pi fk}$$

$$= 1 + \frac{\alpha e^{-j2\pi f}}{1-\alpha e^{-j2\pi f}} + \frac{\alpha e^{j2\pi f}}{1-\alpha e^{j2\pi f}}$$

$$= \frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos 2\pi f}$$

$$S_X(f) = \mathcal{F}\left[4(\alpha)^{|k|} + 16(\beta)^{|k|}\right]$$

$$= 4\frac{1-(\alpha)^2}{1+(\alpha)^2-2(\alpha)\cos 2\pi f} + 16\frac{1-(\beta)^2}{1+(\beta)^2-2(\beta)\cos 2\pi f}$$



10.14

$$\begin{aligned}
S_X(f) &= \sum_{k=-N}^{N}\left(1-\frac{|k|}{N}\right)e^{-j2\pi fk} \\
&= 1+\sum_{k=1}^{N-1}\left(1-\frac{k}{N}\right)e^{-j2\pi fk}+\sum_{k=-(N-1)}^{-1}\left(1+\frac{k}{N}\right)e^{-j2\pi fk} \qquad k'=-k \\
&= 1+\sum_{k=1}^{N-1}\left(1-\frac{k}{N}\right)e^{-j2\pi fk}+\sum_{k'=1}^{N-1}\left(1-\frac{k'}{N}\right)e^{j2\pi fk'}
\end{aligned}$$

We need the following summations

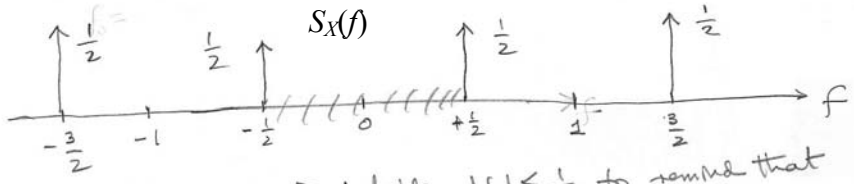
$$\sum_{k=1}^{N-1}\alpha^k = \alpha\sum_{k'=0}^{N-2}\alpha^{k'} = \frac{\alpha-\alpha^N}{1-\alpha}$$

Taking derivatives with respect to α:

$$\begin{aligned}
\sum_{k=1}^{N-1}k\alpha^{k-1} &= \frac{(1-N\alpha^{N-1})(1-\alpha)+(\alpha-\alpha^N)}{(1-\alpha)^2} \\
\Rightarrow \sum_{k=1}^{N-1}k\alpha^{k} &= \frac{(\alpha-N\alpha^{N})(1-\alpha)+\alpha(\alpha-\alpha^N)}{(1-\alpha)^2} \\
\sum_{k=1}^{N-1}\left(1-\frac{k}{N}\right)\alpha^k &= \sum_{k=1}^{N-1}\alpha^k-\frac{1}{N}\sum_{k=1}^{N-1}k\alpha^k \\
&= \frac{\alpha-\alpha^N}{1-\alpha}-\frac{1}{N}\frac{(\alpha-N\alpha^N)(1-\alpha)+\alpha(\alpha-\alpha^N)}{(1-\alpha)^2} \\
&= \left(1-\frac{1}{N}\right)\frac{\alpha}{1-\alpha}-\frac{1}{N}\frac{\alpha^2(1-\alpha^{N-1})}{(1-\alpha)^2} \\
S_X(f) &= 1+\left(1-\frac{1}{N}\right)\left\{\frac{e^{-j2\pi f}}{1-e^{-j2\pi f}}+\frac{e^{j2\pi f}}{1-e^{j2\pi f}}\right\} \\
&\quad -\frac{1}{N}\left\{\left(\frac{e^{-j2\pi f}}{1-e^{-j2\pi f}}\right)^2(1-e^{-j2\pi f(N-1)})+\left(\frac{e^{j2\pi f}}{1-e^{j2\pi f}}\right)^2(1-e^{j2\pi f(N-1)})\right\} \\
&= 1+\left(1-\frac{1}{N}\right)\left\{\frac{e^{-j\pi f}}{e^{j\pi f}-e^{-j\pi f}}+\frac{e^{j\pi f}}{e^{-j\pi f}-e^{+j\pi f}}\right\} \\
&\quad -\frac{1}{N}\left\{\left(\frac{e^{-j\pi f}}{e^{j\pi f}-e^{-j\pi f}}\right)^2 e^{-j\pi f(N-1)}(e^{j\pi f(N-1)}-e^{-j\pi f(N-1)})\right. \\
&\quad \left.+\left(\frac{e^{j\pi f}}{e^{-j\pi f}-e^{j\pi f}}\right)^2 e^{j\pi f(N-1)}(e^{-j\pi f(N-1)}-e^{+j\pi f(N-1)})\right\} \\
&= 1+\left(1-\frac{1}{N}\right)\left\{\frac{e^{-j\pi f}-e^{j\pi f}}{2j\sin\pi f}\right\} \\
&\quad -\frac{1}{N}\left\{e^{-j\pi f(N+1)}\frac{2j\sin\pi f(N-1)}{-4\sin^2\pi f}+e^{j\pi f(N+1)}\frac{(-2j)\sin\pi f(N-1)}{-4\sin^2\pi f}\right\} \\
&= \frac{1}{N}\left\{1-\frac{\sin\pi f(N-1)}{4\sin^2\pi f}\sin\pi f(N+1)\right\} \\
&= \frac{\sin^2\pi Nf}{4N\sin^2\pi f}
\end{aligned}$$

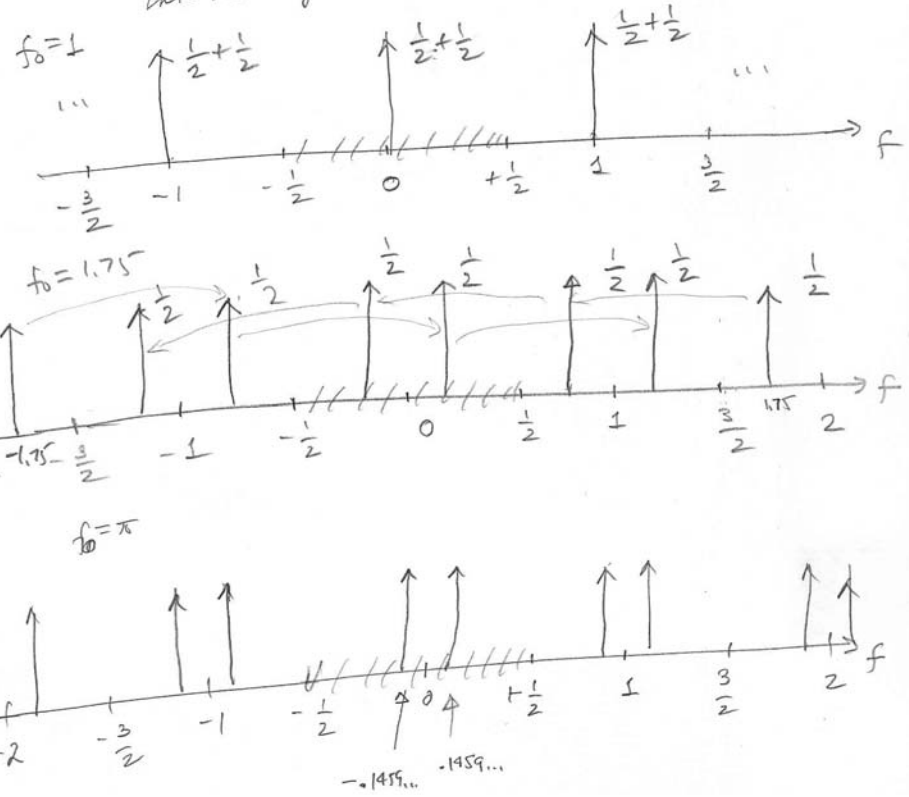
10.15 $X_n = \cos(2\pi f_0 n + \Theta)$

$$\begin{aligned}
R_X(n, n+k) &= \mathcal{E}[X_n X_{n+k}] = \mathcal{E}[\cos(2\pi f_0 n + \Theta)\cos(2\pi f_0(n+k) + \Theta)] \\
&= \mathcal{E}\left[\frac{1}{2}\cos 2\pi f_0 k + \frac{1}{2}\cos(4\pi f_0 n + 2\pi f_0 k + 2\Theta)\right] \\
&= \frac{1}{2}\cos 2\pi f_0 k \\
S_X(f) &= \sum_{k=-\infty}^{\infty} \frac{1}{2}\cos 2\pi f_0 k e^{-j2\pi fk} \\
&= \frac{1}{2}\sum_{k=-\infty}^{\infty} (e^{j2\pi f_0 k} + e^{-j2\pi f_0 k})e^{-j2\pi fk} \\
&= \frac{1}{2}\sum_{k=-\infty}^{\infty} \left\{e^{j2\pi(f+f_0)k} + e^{-j2\pi(f-f_0)k}\right\} \\
&= \frac{1}{2}\delta(f+f_0) + \frac{1}{2}\delta(f-f_0)
\end{aligned}$$



We show frequencies outside $|f| \le \frac{1}{2}$ to remind that the power spectral density of discrete-time processes is periodic.

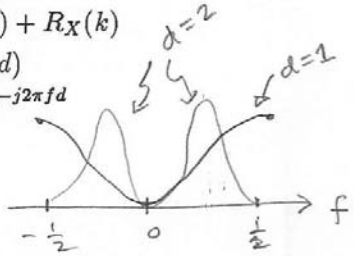
Thus sinusoids with frequencies outside $|f| \le \frac{1}{2}$ reflect back into the range $|f| \le \frac{1}{2}$.



10.16

7.12 a) $\mathcal{E}[D_n] = \mathcal{E}[X_n] - \mathcal{E}[X_{n-d}] = 0$

$$\begin{aligned} R_D(n, n+k) &= \mathcal{E}[(X_n - X_{n-d})(X_{n+k} - X_{n+k-d})] \\ &= R_X(k) - R_X(d+k) - R_X(k-d) + R_X(k) \\ &= 2R_X(k) - R_X(k+d) - R_X(k-d) \\ S_D(f) &= 2S_X(f) - S_X(f)e^{j2\pi fd} - S_X(f)e^{-j2\pi fd} \\ &= 2S_X(f)(1 - \cos 2\pi fd) \end{aligned}$$



b) $\mathcal{E}[D_n^2] = R_D(0) = 2R_X(0) - 2R_X(d)$

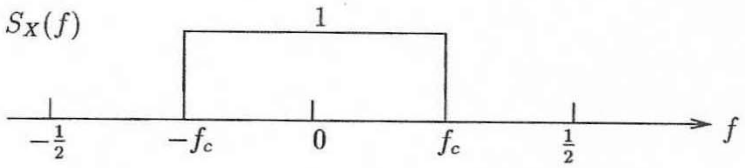
10.17

7.13 The moving average process with $\alpha = 1$ has

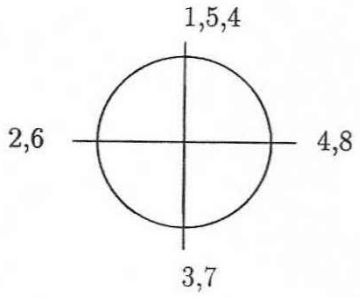
$$R_X(k) = \begin{cases} 2\sigma_X^2 & k = 0 \\ \sigma_X^2 & k = \pm 1 \\ 0 & k > 1 \end{cases}$$

$$\begin{aligned} R_D(k) &= 2R_X(k) - R_X(k+d) - R_X(k-d) \\ S_D(f) &= 2S_X(f)(1 - \cos 2\pi fd) \\ &= 4\sigma_X^2(1 + \cos 2\pi f)(1 - \cos 2\pi fd) \\ &= 4\sigma_X^2(1 - \cos 2\pi fd + \cos 2\pi f - \frac{1}{2}\cos 2\pi f(d-1) \\ &\quad -\frac{1}{2}\cos 2\pi f(d+1)) \end{aligned}$$

10.18



$$R_X(k) = \int_{-f_c}^{f_c} e^{j2\pi fk} df = \frac{e^{j2\pi fk}}{j2\pi k}\Big|_{-f_c}^{f_c} = \frac{e^{j2\pi f_c k} - e^{-j2\pi f_c k}}{j2\pi k}$$
$$= \frac{\sin 2\pi f_c k}{\pi k}$$



If $f_c = \frac{1}{4}$

$$R_X(k) = \frac{\sin k\frac{\pi}{2}}{\pi k} = \begin{cases} \frac{1}{2} & k = 0 \\ \frac{1}{\pi k} & k = 1, 5, 9, \ldots \\ -\frac{1}{\pi k} & k = 3, 7, 11, \ldots \\ 0 & k \text{ even} \end{cases}$$

10.19

7.15 a) $\mathcal{E}[Y_n] = \mathcal{E}[W_n]\mathcal{E}[X_n] = 0$

$$\mathcal{E}[Y_n, Y_{n+k}] = \mathcal{E}[W_n W_{n+k} X_n X_{n+k}]$$
$$= \mathcal{E}[W_n W_{n+k}]\mathcal{E}[X_n X_{n+k}]$$
$$= \delta_{k,0}\mathcal{E}[X_n^2] \Rightarrow Y_n \text{ is a white noise sequence}$$

$$\sigma_{Y_n}^2 = R_Y(0) = \mathcal{E}[X_n^2]$$

b)
$$\mathcal{E}[Y_n] = 0$$
$$R_Y(k) = \begin{cases} \left(\frac{1}{2}\right)^0 = 1 & k = 0 \\ 0 & ew \end{cases}$$

$$f_{Y_{n_1} Y_{n_2} \ldots Y_{n_k}}(y_1, \ldots, y_k) = f_{Y_{n_1}}(y_1) \ldots f_{Y_{n_k}}(y_k)$$
$$= \frac{e^{-\frac{1}{2}(y_1^2 + \ldots + y_k^2)}}{\sqrt{2\pi}^k}$$

10.20 ~~7.16~~

$$\begin{aligned}
\tilde{x}_T(f) &= \int_0^T a\cos(2\pi f_0 t' + \Theta)e^{-j2\pi ft'}dt' \\
&= \frac{a}{2}\int_0^T e^{j\Theta}e^{j2\pi f_0 t'}e^{-j2\pi ft'}dt' + \frac{a}{2}\int_0^T e^{-j\Theta}e^{-j2\pi f_0 t'}e^{-j2\pi ft'}dt' \\
&= \frac{a}{2}e^{j\Theta}\left[\frac{e^{j2\pi(f_0-f)T}-1}{j2\pi(f_0-f)}\right] + \frac{a}{2}e^{-j\Theta}\left[\frac{1-e^{-j2\pi(f_0+f)T}}{j2\pi(f_0+f)}\right] \\
&= \frac{a}{2}e^{j\Theta}e^{j\pi(f_0-f)T}\left[\frac{e^{j\pi(f_0-f)T}-e^{-j\pi(f_0-f)T}}{j2\pi(f_0-f)}\right] \\
&\quad + \frac{a}{2}e^{-j\Theta}e^{-j\pi(f_0+f)}\left[\frac{e^{j\pi(f_0+f)T}-e^{-j\pi(f_0+f)T}}{j2\pi(f_0+f)}\right] \\
&= \frac{a}{2}e^{j\Theta}e^{j\pi(f_0-f)}\frac{\sin\pi(f_0-f)T}{\pi(f_0-f)} \\
&\quad + \frac{a}{2}e^{-j\Theta}e^{-j\pi(f_0+f)}\frac{\sin\pi(f_0+f)T}{\pi(f_0+f)}
\end{aligned}$$

$$\begin{aligned}
\tilde{p}_T(f) &= \frac{1}{T}|\tilde{x}_T(f)|^2 \\
&= \frac{a^2}{4T}\frac{\sin^2\pi(f_0-f)T}{\pi(f_0-f)^2} + \frac{a^2}{4T}\frac{\sin^2\pi(f_0+f)T}{\pi^2(f_0+f)^2}
\end{aligned}$$

$$\begin{aligned}
&+\frac{a^2}{4T}e^{j2\Theta}e^{j\pi(2f_0)}\frac{\sin\pi(f_0-f)T}{\pi(f_0-f)}\frac{\sin\pi(f_0+f)T}{\pi(f_0+f)} \\
&+\frac{a^2}{4T}e^{-j2\Theta}e^{-j\pi 2f_0}\frac{\sin\pi(f_0-f)T}{\pi(f_0-f)}\frac{\sin\pi(f_0+f)T}{\pi(f_0+f)}
\end{aligned}$$

$$\begin{aligned}
\tilde{p}_T(f) &= \frac{a^2}{4T}\frac{\sin^2\pi(f_0-f)T}{\pi^2(f_0-f)^2} + \frac{a^2}{4T}\frac{\sin^2\pi(f_0+f)T}{\pi^2(f_0+f)^2} \\
&\quad + \frac{a^2}{4T}\cos(2\pi f_0 + 2\Theta)\frac{\sin\pi(f_0-f)T}{\pi(f_0-f)}\frac{\sin\pi(f_0+f)T}{\pi(f_0+f)}
\end{aligned}$$

NOTE

$$\begin{aligned}
\tilde{p}_T(f_0) &= \frac{a^2T}{4} + \frac{a^2}{4T}\frac{\sin^2\pi 2f_0T}{\pi^2 4f_0^2} + \frac{a^2\cos(2\pi f_0+2\Theta)}{4}\frac{\sin\pi(2f_0T)}{2\pi f_0} \\
\tilde{p}_T(-f_0) &= \frac{a^2T}{4} + \frac{a^2}{4T}\frac{\sin^2\pi 2f_0T}{\pi^2 4f_0^2} + \frac{a^2}{4}\cos(2\pi f_0+2\Theta)\frac{\sin 2\pi f_0T}{2\pi f_0}
\end{aligned}$$

$\therefore$ as $T \to \infty$

$$\tilde{p}_T(f) \to \frac{a^2}{4}\delta(f-f_0) + \frac{a^2}{4}\delta(f+f_0)$$

10.21

```
%P.10.21
clear all;
close all;
samples = 50;
len=128;
x=randn(len,samples);
xf=fft(x);
pf=sqrt(xf'.*conj(xf'))/len;
sf=mean(pf);
sf1=mean(pf(1:10,:));
sf2=mean(pf(1:20,:));
sf3=mean(pf(1:30,:));
sf4=mean(pf(1:40,:));
subplot(2,3,1);
plot(1:len,sf);
axis([1 len 0.06 0.11]);
title('All (50) realizations');
subplot(2,3,2);
plot(1:len,sf1);
axis([1 len 0.06 0.11]);
title('10 realizations');
subplot(2,3,3);
plot(1:len,sf2);
axis([1 len 0.06 0.11]);
title('20 realizations');
subplot(2,3,4);
plot(1:len,sf3);
axis([1 len 0.06 0.11]);
title('30 realizations');
subplot(2,3,5);
plot(1:len,sf4);
axis([1 len 0.06 0.11]);
title('40 realizations');
```

10.2 Response of Linear Systems to Random Signals

10.22

a) $S_Y(f) = |H(f)|^2 S_X(f) = 4\pi^2 f^2 S_X(f)$

b) $R_Y(\tau) = \mathcal{F}^{-1}[S_Y(f)] = -\frac{d^2}{d\tau^2} R_X(\tau)$

10.23

a) $S_Y(f) = 4\pi^2 f^2 \frac{N_0}{2} \qquad \mathcal{F} < W$

$$\begin{aligned} R_Y(\tau) &= \int_{-W}^{W} 2\pi^2 N_0 f^2 e^{j2\pi f\tau} df \\ &= 2\pi^2 N_0 \left[e^{j2\pi f\tau} \frac{(-4\pi^2 f^2\tau^2 - 2j2\pi f\tau + 2)}{(j2\pi\tau)^3} \right]_{-W}^{W} \\ &= 2\pi^2 N_0 \left[e^{j2\pi W\tau} \frac{(2 - 4\pi^2 W^2\tau^2 - 4j\pi W\tau)}{-j8\pi^3\tau^3} \right. \\ &\qquad \left. - e^{-j2\pi W\tau} \frac{(2 - 4\pi^2 W^2\tau^2 + 4j\pi W\tau)}{-j8\pi^3\tau^3} \right] \\ &= \frac{4\pi^2 N_0}{8\pi^3\tau^3} [-(2 - 4\pi^2 W^2\tau^2)\sin 2\pi W\tau + 4\pi W\tau \cos 2\pi W\tau] \end{aligned}$$

b) $R_Y(0) = \int_{-W}^{W} S_Y(f) df = \int_{-W}^{W} 4\pi^2 f^2 \frac{N_0}{2} = \frac{4\pi^2 N_0 W^3}{3}$

10.24

$$S_X(f) = \beta^2 e^{-\pi f^2} \Rightarrow R_X(\tau) = \beta^2 e^{-\pi\tau^2}$$

$$S_Y(f) = 4\pi^2 f^2 \beta^2 e^{-f^2/2}$$

$$R_Y(\tau) = -\frac{d^2}{d\tau^2} R_X(\tau)$$

$$= -\frac{d}{d\tau}\left[\beta^2 e^{-\pi\tau^2}(-2\pi\tau)\right]$$

$$= -\beta^2\left[e^{-\pi\tau^2}(-2\pi\tau) - e^{-\pi\tau^2}(-2\pi)\right]$$

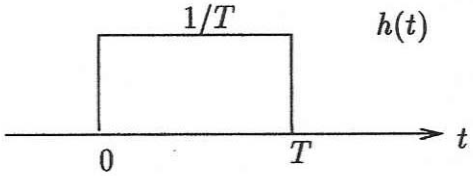
$$= 2\pi\beta^2 e^{-\pi\tau^2}\left[\tau - 1\right]$$

10.25

7.19 a) The impulse response is

$$h(t) = \frac{1}{T}\int_{t-T}^{t}\delta(t')dt' = \frac{1}{T}\int_{-\infty}^{t}\delta(t')dt' - \frac{1}{T}\int_{-\infty}^{t-T}\delta(t')dt'$$

$$= \frac{1}{T}[u(t) - u(t-T)]$$



$$H(f) = \frac{1}{T}\int_0^T e^{-j2\pi ft}dt = \frac{1}{T}\frac{1-e^{-j2\pi ft}}{j2\pi f} = \frac{1}{T}\frac{e^{j2\pi f\frac{T}{2}} - e^{-j2\pi f\frac{T}{2}}}{j2\pi f}e^{-j2\pi f\frac{T}{2}}$$

$$= \frac{1}{T}\frac{\sin\pi fT}{\pi f}e^{-j\pi fT}$$

b)

$$S_Y(f) = |H(f)|^2 S_X(f)$$

$$= \frac{\sin^2\pi fT}{T^2\pi^2 f^2}S_X(f)$$

10.26

$$R_X(\tau) = \begin{cases} 1 - \frac{|\tau|}{T} & |\tau| < T \\ 0 & ew \end{cases}$$

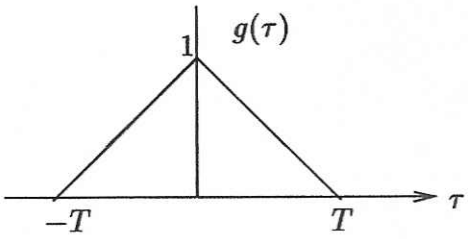
$$\begin{aligned} S_X(f) &= T\left(\frac{\sin \pi fT}{\pi fT}\right)^2 = \frac{1}{T}\left(\frac{\sin \pi fT}{\pi f}\right)^2 \\ S_Y(f) &= \frac{1}{T^3}\left(\frac{\sin \pi fT}{\pi f}\right)^4 \\ \mathcal{E}[Y^2(t)] &= R_Y(0) \end{aligned}$$

$$S_Y(f) = TG(f)G(f)$$

where

$$G(f) = T\left(\frac{\sin \pi fT}{\pi fT}\right)^2$$

and



$$\begin{aligned} \therefore R_Y(\tau) &= Tg(\tau) \star g(\tau) \\ &= T\int_{-\infty}^{\infty} g(\lambda)g(\tau - \lambda)d\lambda \end{aligned}$$

and

$$\begin{aligned} \mathcal{E}[Y^2(t)] &= R_Y(0) = T\int_{-\infty}^{\infty} g(\lambda)g(-\lambda)d\lambda \\ &= T\int_{-T}^{T} g^2(\lambda)d\lambda \quad \text{since } g(\lambda) = g(-\lambda) \\ &= 2T\int_{0}^{T}\left(1 - \frac{\lambda}{T}\right)^2 d\lambda \\ &= \frac{2T^2}{3} \end{aligned}$$

10.27

7.21 **a)** $$S_{YX}(f) = H(f)S_X(f) = \frac{N_0/2}{1+j2\pi f}$$

$$R_{YX} = \mathcal{F}^{-1}[S_{YX}(f)] = \frac{N_0}{2}e^{-\tau} \qquad \tau > 0$$

b) $$S_Y(f) = |H(f)|^2 S_X(f) = \frac{N_0/2}{1+4\pi^2 f^2}$$

$$R_Y(\tau) = \mathcal{F}^{-1}[S_Y(f)] = \frac{N_0}{4}e^{-|\tau|}$$

c) $$R_Y(0) = \frac{N_0}{4}$$

10.28

a)

$$S_{YX}(f) = H(f)S_X(f) = (1+j2\pi f)S_X(f) = S_X(f) + j2\pi f S_X(f)$$

$$R_{YX}(\tau) = \mathcal{F}^{-1}\{S_{YX}(f)\} = R_X(\tau) + R_X'(\tau)$$

b)

$$S_Y(f) = |H(f)|^2 S_X(f) = (1+(2\pi f)^2)S_X(f) = S_X(f) + (2\pi f)^2 S_X(f)$$

$$R_Y(\tau) = \mathcal{F}^{-1}\{S_Y(f)\} = R_X(\tau) - R_X''(\tau)$$

c)

average power: $R_Y(0) = R_X(0) - R_X''(0)$

10.29

~~7.22~~

$$\begin{aligned} Y(t) &= \int_0^\infty h(t-s)X(s)ds = \int_{-\infty}^t h(u)X(t-u)du \\ \mathcal{E}[Y(t)] &= \int_{-\infty}^t h(u)\mathcal{E}[X(t-u)]du = m_X \int_{-\infty}^t h(u)du \\ \mathcal{E}[Y(t)Y(t+\tau)] &= \mathcal{E}\left[\int_{-\infty}^t \int_{-\infty}^{t+\tau} h(u)h(v)X(t-u)X(t+\tau-v)dudv\right] \\ &= \int_{-\infty}^t \int_{-\infty}^{t+\tau} h(u)h(v)R_X(\tau+u-v)dudv \end{aligned}$$

depends on t and $t+\tau$. As $t \to \infty$ this expression approaches the expression in Eqn 10.42 and the process becomes WSS.

10.30

~~7.23~~ $X(t) \to h(t) \to Y(t)$

From Eqn. 10.45a:

$$\begin{aligned} S_{YX}(f) &= H(f)S_X(f) = H(f)\frac{N_0}{2} \\ &\qquad \text{input is white noise} \\ \Rightarrow R_{YX}(\tau) &= \mathcal{F}^{-1}[S_{YX}(f)] = \frac{N_0}{2}h(t) \quad \surd \end{aligned}$$

10.31 ~~7.24~~

$$Y(t_1) = \int_{-\infty}^{\infty} h_1(u)X(t_1-u)du \approx \sum_i h_i(u_i)X(t_1-x_i)\Delta u$$
$$W(t_2) = \int_{-\infty}^{\infty} h_2(v)X(t_2-v)dv \approx \sum_j h_2(v_j)X(t_2-v_j)\Delta v$$

$\therefore Y(t_1)$ and $W(t_2)$ are jointly Gaussian

$$\mathcal{E}[Y(t_1)W(t_2)] = \mathcal{E}\left[\int_{-\infty}^{\infty} h_1(u)X(t_1-u)du \int_{-\infty}^{\infty} h_2(v)X(t_2-v)dv\right]$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_1(u)h_2(v)R_X(\underbrace{t_2-t_1}_{\tau}-v+u)dudv$$

$\therefore Y(t_1)$ and $W(t_2)$ are jointly WSS.

From Eqn. 10.42, $Y(t)$ has variance function

$$\sigma_Y^2(t) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_1(s)h_1(r)R_X(s-r)dsdr - m_X^2 H_1(0)^2$$

and $W(t)$ has

$$\sigma_W^2(t) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_2(s)h_2(r)R_X(s-r)dsdr - m_X^2 H_2(0)^2$$

If $X(t)$ is white noise then

$$R_X(\tau) = \sigma_X^2 \delta(\tau)$$

$$E[Y(t_1)W(t_2)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_1(u)\, h_2(v)\, \delta_X(\tau - v + u)\, du\, dv$$

$$= \int_{-\infty}^{\infty} h_1(u)\, h_2(u+\tau)\, du$$

correlation of impulse responses

$$\sigma_Y^2(t) = \int\int_{-\infty}^{\infty} h_1(s)\, h_1(r)\, \delta_X(s-r)\, ds\, dr - m_X^2 H_1(0)^2$$

$$= \int_{-\infty}^{\infty} h_1^2(s)\, ds - m_X^2 H_1^2(0)$$

10.32

from 10.31:

$$E[Y(t_1)W(t_2)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_1(u)\, h_2(v)\, R_X(t_2 - t_1 - v + u)\, du\, dv$$

$$S_{YW}(f) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_1(u)\, h_2(v)\, R_X(\tau - v + u)\, du\, dv\, e^{-j2\pi f \tau}\, d\tau$$

$\tau' = \tau - v + u$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_1(u)\, h_2(v)\, R_X(\tau')\, du\, dv\, e^{-j2\pi f(\tau' + v - u)}\, d\tau'$$

$$= \int_{-\infty}^{\infty} h_1(u)\, e^{j2\pi f u}\, du \int_{-\infty}^{\infty} h_2(v)\, e^{-j2\pi f v}\, dv \int_{-\infty}^{\infty} R_X(\tau')\, e^{-j2\pi f \tau'}\, d\tau'$$

$$= H_1^*(f)\, H_2(f)\, S_X(f)$$

$= 0$ if $H_1(f)$ & $H_2(f)$ are non-overlapping.

$\Rightarrow R_{YW}(t_1, t_2) = 0$

$\Rightarrow$ $Y(t_1)$ and $W(t_2)$ are uncorrelated.

$\Rightarrow$ $Y(t_1)$ and $W(t_2)$ are independent since they are jointly Gaussian

10.33

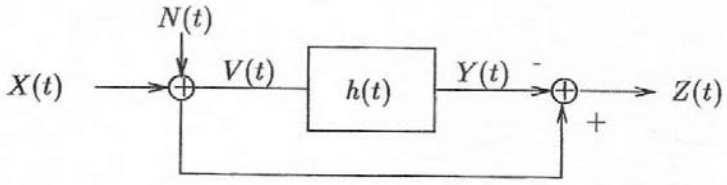
~~7.25~~ $Z(t) = X(t) - Y(t) \qquad Y(t) = h(t) \star X(t)$

a)
$$\begin{aligned} E[Z(t)Z(t+\tau)] &= \mathcal{E}[(X(t)-Y(t))(X(t+\tau)-Y(t+\tau))] \\ &= R_X(\tau) + R_Y(\tau) - R_{YX}(-\tau) - R_{XY}(-\tau) \\ &= R_X(\tau) + R_Y(\tau) - R_{XY}(\tau) - R_{XY}(-\tau) \\ S_Z(f) &= S_X(f) + S_Y(f) - S_{XY}(f) - S_{XY}^*(f) \\ &= S_X(f) + |H(f)|^2 S_X(f) - H^*(f)S_X(f) - H(f)S_X(f) \\ &= \left\{ 1 + |H(f)|^2 - \underbrace{(H^*(f) + H(f))}_{2\mathrm{Re}[H(f)]} S_X(f) \right\} \\ &= |\{1 - H(f)\}|^2 S_X(f) \end{aligned}$$

b)
$$\begin{aligned} \mathcal{E}[Z(t)^2] &= R_X(0) + R_Y(0) - 2R_{XY}(0) \\ &= \mathcal{E}[X^2(t)] + \int\int_{-\infty}^{\infty} h(s)h(r)R_X(s-r)dsdr \\ &\quad -2\int_{-\infty}^{\infty} h(r)R_X(-r)dr \end{aligned}$$

Also
$$\mathcal{E}[Z^2(t)] = \int_{-\infty}^{\infty} |1 - H(f)|^2 S_X(f)df$$

10.34



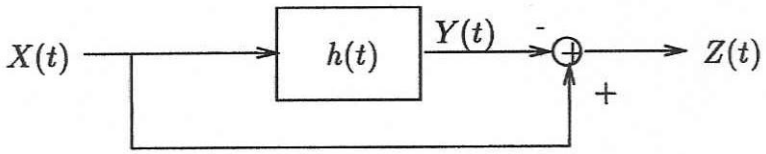
$$
\begin{aligned}
R_Z(\tau) &= \mathcal{E}[Z(t+\tau)Z(t)] = \mathcal{E}[(X(t+\tau)-Y(t+\tau))(X(t)-Y(t))] \\
&= R_X(\tau)+R_Y(\tau)-R_{XY}(\tau)-R_{YX}(\tau) \\
R_{XY}(\tau) &= \mathcal{E}[X(t+\tau)Y(t)] = \mathcal{E}\left[X(t+\tau)\int_{-\infty}^{\infty} h(\lambda)V(t-\lambda)d\lambda\right] \\
&= \int_{-\infty}^{\infty} h(\lambda)R_{XV}(\tau+\lambda)d\lambda \\
&= \int_{-\infty}^{\infty} h(\lambda)R_X(\tau+\lambda)d\lambda \quad \text{since } R_{XV}(\tau) = \mathcal{E}[X(t+\tau)(X(t)+N(t))] = R_X(\tau) \\
&= h(-\tau)\star R_X(\tau) \\
S_Z(f) &= S_X(f)+S_Y(f)-S_{XY}(f)-S_{YX}(f) \\
&= S_X(f)+|H(f)|^2(S_X(f)+S_N(f))-H(f)S_X(f)-H^*(f)S_X(f) \\
S_Z(f) &= |1-H(f)|^2S_X(f)+|H(f)|^2S_N(f) \qquad (*)
\end{aligned}
$$

Comments: If we view $Y(t)$ as our estimate for $X(t)$, then $S_Z(f)$ is the power spectral density of the error signal $Z(t) = Y(t) - X(t)$. Equation (*) suggests the following:

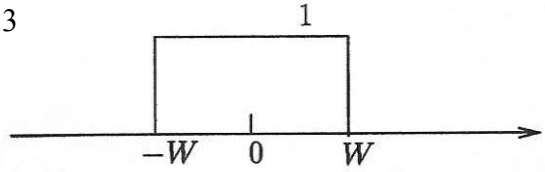
if $S_X(f) >> S_N(f)$ let $H(f) \approx 1$
if $S_X(f) << S_N(f)$ let $H(f) \approx 0$

i.e. select $H(f)$ to "pass" the signal and reject the noise.

10.35



From Problem 10.33



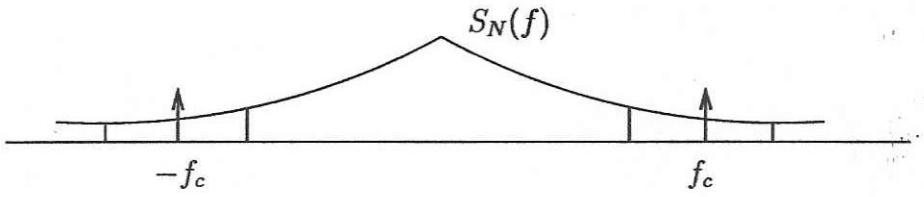
$$
\begin{aligned}
S_Z(f) &= |1 - H(f)|^2 S_X(f) \\
&= \begin{cases} 0 & |f| < W \\ \frac{4\alpha}{4\alpha^2 + 4\pi^2 f^2} & |f| > W \end{cases} \\
\mathcal{E}[Z^2(t)] &= 2\int_W^\infty \frac{4\alpha}{4\alpha^2 + 4\pi^2 f^2} df \qquad \begin{array}{l} x = 2\pi f \\ dx = 2\pi df \end{array} \\
&= 8\alpha \int_{2\pi W}^\infty \frac{1}{4\alpha^2 + x^2} \frac{dx}{2\pi} \\
&= \frac{4\alpha}{\pi} \frac{1}{2\alpha} \tan^{-1} \frac{x}{2\alpha}\Big|_{2\pi W}^\infty \\
&= \frac{2}{\pi}\left[\frac{\pi}{2} - \tan^{-1}\frac{\pi W}{\alpha}\right] \\
&= 1 - \frac{2}{\pi}\tan^{-1}\frac{\pi W}{\alpha}
\end{aligned}
$$

10.36

~~7.28~~ $Y(t) = a\cos(\omega_c(t) + \Theta) + N(t)$

We will assume that Θ and $N(t)$ are statistically independent:

$$\begin{aligned} \therefore R_Y(\tau) &= a\cos\omega_c\tau + R_N(\tau) \\ S_Y(f) &= \frac{a^2}{4}\delta(f-f_c) + \frac{a^2}{4}\delta(f+f_c) + S_N(f) \end{aligned}$$



$$\begin{aligned} \text{Signal Power} &= \int_{-f_c-W}^{-f_c+W}\left(\frac{a^2}{4}\delta(f-f_c) + \frac{a^2}{4}\delta(f+f_c)\right)df \\ &\quad + \int_{f_c-W}^{f_c+W}\left(\frac{a^4}{4}\delta(f-f_c) + \frac{a^2}{4}\delta(f+f_c)\right)df \\ &= \frac{a^2}{4} + \frac{a^2}{4} = \frac{a^2}{2} \end{aligned}$$

$$\begin{aligned} \text{Noise Power} &= \int_{-f_c-W}^{-f_c+W} S_N(f)df + \int_{f_c-W}^{f_c+W} S_N(f)df \\ &= 2\int_{f_c-W}^{f_c+W} S_N(f)df \\ \text{SNR} &= \frac{a^2/2}{2\int_{f_c-W}^{f_c+W} S_N(f)df} \end{aligned}$$

10.37

7.31 Find the impulse response:

If input is $\delta_n = 1$, $n = 0$ and $\delta_n = 0$ elsewhere

$$\begin{aligned}
h_n &= \begin{cases} \frac{1}{3} & n = -1, 0, 1 \\ 0 & \text{elsewhere} \end{cases} \\
\Rightarrow H(f) &= \frac{1}{3}[e^{-j2\pi f} + 1 + e^{j2\pi f}] \\
&= \frac{1}{3}[1 + 2\cos 2\pi f] \\
S_Y(f) &= |H(f)|^2 S_X(f) = \frac{1}{9}[1 + 2\cos 2\pi f]^2 S_X(f) \\
&= \frac{1}{9}[1 + e^{j2\pi f} + e^{-j2\pi f}]^2 S_X(f) \\
&= \frac{1}{9}[1 + e^{j2\pi f} + e^{-j2\pi f} + e^{j2\pi f} + e^{-j2\pi f} \\
&\quad +1 + e^{-j2\pi f} + 1 + e^{-j4\pi f}] S_X(f) \\
&= \frac{1}{9}[3 + 2e^{j2\pi f} + 2e^{-j2\pi f} + e^{j4\pi f} + e^{-j4\pi f}] S_X(f) \\
\therefore R_Y(k) &= \mathcal{F}^{-1}[S_Y(f)] = \frac{1}{3}R_X(k) + \frac{2}{9}R_X(k+1) + \frac{2}{9}R_X(k-1) \\
&\quad + \frac{1}{9}R_X(k+2) + \frac{1}{9}R_X(k-2) \\
\mathcal{E}[Y_n^2] &= \frac{1}{3}R_X(0) + \frac{4}{9}R_X(1) + \frac{2}{9}R_X(2)
\end{aligned}$$

10.38

7.32 $R_X(n) = \begin{cases} \sigma^2 & n = 0 \\ 0 & n \neq 0 \end{cases}$ where we assumed $\mathcal{E}[X_n] = 0$

$$\Rightarrow R_Y(k) = \begin{cases} \frac{1}{3}\sigma^2 & k = 0 \\ \frac{2}{9}\sigma^2 & k = \pm 1 \\ \frac{1}{9}\sigma^2 & k = \pm 2 \\ 0 & ew \end{cases}$$

$\therefore (Y_n, Y_{n+1}, Y_{n+2})$ is jointly Gaussian with zero mean vector and covariance matrix

$$K = \sigma^2 \begin{bmatrix} \frac{1}{3} & \frac{2}{9} & \frac{1}{9} \\ \frac{2}{9} & \frac{1}{3} & \frac{2}{9} \\ \frac{1}{9} & \frac{2}{9} & \frac{1}{3} \end{bmatrix}$$

10.39
~~7.30~~ a)

$$\begin{aligned} R_{YX}(k) &= \mathcal{E}[Y_{n+k}X_n] = \mathcal{E}[(X_{n+k} + \beta X_{n+k-1})X_n] \\ &= R_X(k) + \beta R_X(k-1) \\ S_{YX}(f) &= \mathcal{F}[R_{YX}(k)] = S_X(f) + \beta S_X(f)e^{-j2\pi f} \\ &= \frac{(1+\beta e^{-j2\pi f})(1-\alpha^2)}{1+\alpha^2-2\alpha\cos 2\pi f}\sigma^2 \end{aligned}$$

since

$$\mathcal{F}[\alpha^{|k|}] = \frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos 2\pi f}$$

See Problem ~~7.9~~ 10.13.

b)

$$\begin{aligned} R_Y(k) &= \mathcal{E}[(X_{n+k} + \beta X_{n+k-1})(X_n + \beta X_{n-1})] \\ &= (1+\beta^2)R_X(k) + \beta R_X(k+1) + \beta R_X(k-1) \\ S_Y(f) &= (1+\beta^2)S_X(f) + \beta S_X(f)e^{j2\pi f} + \beta S_X(f)e^{-j2\pi f} \\ &= [(1+\beta^2) + 2\beta\cos 2\pi f]S_X(f) \\ &= \frac{1+\beta^2+2\beta\cos 2\pi f}{1+\alpha^2-2\alpha\cos 2\pi f}(1-\alpha^2)\sigma^2 \\ \mathcal{E}[Y_n^2] &= R_Y(0) = (1+\beta^2)R_X(0) + \beta R_X(1) + \beta R_X(-1) \\ &= (1+\beta^2)\sigma^2 + 2\beta\sigma^2\alpha \end{aligned}$$

c) if $\beta = -\alpha$ then $S_Y(f) = (1-\alpha^2)\sigma^2$ and $\mathcal{E}[Y_n^2] = (1-\alpha^2)\sigma^2$.

10.40

~~29 a)~~

$$H(f) = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^n e^{-j2\pi fn} = \frac{1}{1-\frac{1}{2}e^{-j2\pi f}}$$

$$G(f) = \sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^n e^{-j2\pi fn} = \frac{1}{1-\frac{1}{4}e^{-j2\pi f}}$$

$$S_Y(f) = |H(f)|^2 \frac{N_0}{2} = \frac{N_0/2}{\frac{5}{4}-\cos 2\pi f}$$

$$S_Z(f) = |G(f)|^2 S_Y(f) = \frac{N_0/2}{\left(\frac{5}{4}-\cos 2\pi f\right)\left(\frac{17}{16}-\frac{1}{2}\cos 2\pi f\right)}$$

b)

$$S_{WY}(f) = H(f)S_X(f) = \frac{N_0/2}{1-\frac{1}{2}e^{-j2\pi f}}$$

$$\Rightarrow R_{WY}(k) = \frac{N_0}{2}\left(\frac{1}{2}\right)^k u(k)$$

$$S_{WZ}(f) = H(f)G(f)S_X(f) = \frac{N_0/2}{(1-\frac{1}{2}e^{-j2\pi f})(1-\frac{1}{4}e^{-j2\pi f})}$$

$$= \frac{N_0}{1-\frac{1}{2}e^{-j2\pi f}} - \frac{N_0/2}{1-\frac{1}{4}e^{-j2\pi f}}$$

$$R_{WZ}(k) = \left(N_0\left(\frac{1}{2}\right)^k - \frac{N_0}{2}\left(\frac{1}{4}\right)^k\right)u(k)$$

c) $S_Z(f) = \dfrac{N_0/2}{\left(\frac{5}{4}-\cos 2\pi f\right)\left(\frac{17}{16}-\frac{1}{2}\cos 2\pi f\right)} = \dfrac{\frac{8}{7}N_0}{\frac{5}{4}-\cos 2\pi f} - \dfrac{\frac{4}{7}N_0}{\frac{17}{16}-\frac{1}{2}\cos 2\pi f}$

From Problem 10.13 we know that

$$\mathcal{F}[|\alpha|^k] = \sum_{k=-\infty}^{\infty} |\alpha|^k e^{-j2\pi fk} = \frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos 2\pi f}$$

10.41 a)

$$\begin{aligned}\mathcal{E}[X_n X_{n+k}] &= \mathcal{E}\left[\left(\sum_{i=0}^{p}\alpha_i W_{n-i}\right)\left(\sum_{j=0}^{p}\alpha_j W_{n+k-j}\right)\right] \\ &= \sum_{i=0}^{p}\sum_{j=0}^{p}\alpha_i\alpha_j \underbrace{\mathcal{E}[W_{n-i}W_{n+k-j}]}_{R_W(k-j+i)}\end{aligned}$$

$$\begin{aligned}k > p &\Rightarrow k-j+i > 0 \\ &\Rightarrow R_W(k-j+i) = 0 \quad \text{since } W \text{ is white}\end{aligned}$$

Similarly

$$k < p \Rightarrow R_W(k-j+i) = 0$$

$$\therefore \mathcal{E}[X_n X_{n+k}] = 0 \quad \text{for} \quad |k| > p$$

For $|k| \le p$

$$\begin{aligned}R_X(k) &= \sigma^2 \sum_{i=0}^{p}\sum_{l=0}^{p}\alpha_i\alpha_l \overbrace{\delta_{k-l+i}}^{R_W(k-l+i)} \\ &= \sigma^2 \sum_{l=0}^{p}\alpha_l\alpha_{l-k} \quad \text{since } \delta_{k-l+i} = 1\cdot\sigma^2 \Leftrightarrow i = l-k \\ &= \sigma^2 \sum_{l=-\infty}^{\infty}\alpha_l\alpha_{l-k} \quad \text{where we define } \alpha_l = 0,\ l<0 \text{ and } l>p \\ S_X(f) &= \sum_{k=-\infty}^{\infty} R_X(k)e^{-j2\pi fk} \\ &= \sigma^2 \sum_{k=-\infty}^{\infty}\sum_{l=-\infty}^{\infty}\alpha_l\alpha_{l-k}e^{-j2\pi fk} \quad m = l-k \\ &= \sum_{l=-\infty}^{\infty}\alpha_l e^{-j2\pi fl}\sum_{m=-\infty}^{\infty}\alpha_m e^{j2\pi fm} \\ &= H(f)H^*(f)\sigma^2\end{aligned}$$

where $H(f) = \sum_{l=-\infty}^{\infty}\alpha_l e^{-j2\pi fl}$ is the impulse response of the system.

10.42

7.34 a) If input is δ_n then $Y_0 = 1$ and $Y_1 = \frac{3}{4}$. We seek a solution to

$$Y_n = \frac{3}{4}Y_{n-1} - \frac{1}{8}Y_{n-2}$$

of the form

$$Y_n = c_1 z_1^n + c_2 z_2^n$$

that satisfies the above boundary conditions. The z_i must satisfy

$$\begin{aligned} cz^n &= \frac{3}{4}cz^{n-1} - \frac{1}{8}cz^{n-2} \Rightarrow z^2 - \frac{3}{4}z + \frac{1}{8} = 0 \\ &\Rightarrow z_1 = \frac{1}{2} \quad z_2 = \frac{1}{4} \end{aligned}$$

$$\begin{array}{ll} \text{Boundary} & \Rightarrow \quad Y_0 = 1 = c_1 + c_2 \\ \text{Conditions} & \Rightarrow \quad Y_1 = \frac{3}{4} = \frac{c_1}{2} + \frac{c_2}{4} \end{array} \Bigg\} \quad \begin{array}{l} c_1 = 2 \\ c_2 = -1 \end{array}$$

$$\Rightarrow Y_n = 2\left(\frac{1}{2}\right)^n - \left(\frac{1}{4}\right)^n \qquad n \geq 0$$

b)

$$\begin{aligned} H(f) &= 2\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n e^{-j2\pi fn} - \sum_{n=0}^{\infty}\left(\frac{1}{4}\right)^n e^{-j2\pi fn} \\ &= 2\frac{1}{1-\frac{1}{2}e^{-j2\pi f}} - \frac{1}{1-\frac{1}{4}e^{-j2\pi f}} \\ &= \frac{\frac{1}{2}e^{-j2\pi f}}{(1-\frac{1}{2}e^{-j2\pi f})(1-\frac{1}{4}e^{-j2\pi f})} \end{aligned}$$

c)

$$\begin{aligned} S_Y(f) &= |H(f)|^2\sigma_W^2 \\ &= \frac{\sigma_W^2/4}{(\frac{3}{4}-\cos 2\pi f)(\frac{17}{16}-\frac{1}{2}\cos 2\pi f)} \\ S_Y(f) &= \frac{4}{7}\left(\frac{4}{3}\right)\frac{\frac{3}{4}\sigma_X^2}{\frac{5}{4}-\cos 2\pi f} - \frac{2}{7}\frac{16}{15}\frac{\frac{15}{16}\sigma_X^2}{\frac{17}{16}-\frac{1}{2}\cos 2\pi f} \end{aligned}$$

$$\Rightarrow R_Y(k) = \frac{16}{21}\left(\frac{1}{2}\right)^{|k|} - \frac{32}{105}\left(\frac{1}{4}\right)^{|k|}$$

10.13

See Problem 7.8 solution.

10.43

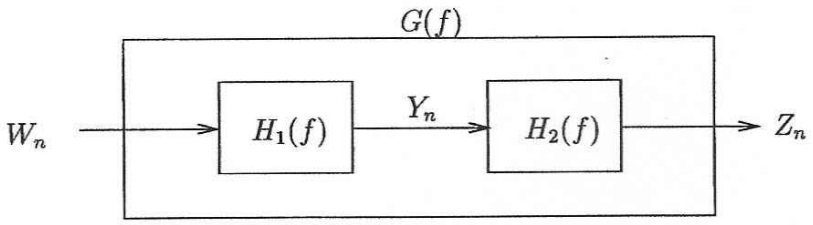
~~7.35~~ $Z_n = Y_n - \frac{1}{4}Y_{n-1}$

a) The impulse response for this system is

$$h_0 = 1 \quad h_1 = -\frac{1}{4} \quad h_n = 0 \quad n \neq 0 \text{ or } 1$$

$$\therefore H_2(f) = 1 - \frac{1}{4}e^{-j2\pi f}$$

Let W_n be the input to the system in Problem 10.42, then



where

$$H_1(f) = \frac{\frac{1}{2}e^{-j2\pi f}}{(1 - \frac{1}{2}e^{-j2\pi f})(1 - \frac{1}{4}e^{-j2\pi f})}$$

The power spectral density of Z_n is

$$\begin{aligned} S_Z(f) &= |G(f)|^2 S_W(f) = |H_1(f)H_2(f)|^2 \sigma_W^2 \\ &= \frac{\sigma_W^2/4}{\frac{5}{4} - \cos 2\pi f} \end{aligned}$$

where $G(f)$ is the transfer function that defines a first-order autoregressive process.

$$R_Z(k) = \frac{\sigma_W^2}{4}\frac{4}{3}\mathcal{F}^{-1}\left[\frac{\frac{3}{4}}{\frac{5}{4} - \cos 2\pi f}\right] = \frac{\sigma_W^2}{3}\left(\frac{1}{2}\right)^{|k|}$$

b) $G(f) = H_1(f)H_2(f) = \frac{\frac{1}{2}e^{-j2\pi f}}{1 - \frac{1}{2}e^{-j2\pi f}} = \frac{1}{2}e^{-j2\pi f}\sum_{l=0}^{\infty}\left(\frac{1}{4}e^{-j2\pi f}\right)$ which corresponds to a first-order autoregressive process.

c) If we let $H_3(f) = 1 - \frac{1}{2}e^{-j2\pi f}$, then

$$|H_1(f)H_2(f)H_3(f)|^2 = \frac{1}{4}$$

and

$$|H_3(f)|^2 S_Z(f) = \frac{\sigma_W^2}{4}$$

10.44

7.36 a)

$$\begin{aligned}\mathcal{E}[Y_n^2] &= \mathcal{E}\left[Y_n\left(\sum_{i=1}^{q}\alpha_i Y_{n-i}+W_n\right)\right]\\ &= \sum_{i=1}^{q}\alpha_i R_Y(i)+R_{YW}(0)\end{aligned}$$

$$\begin{aligned}R_{YW}(0) &= \mathcal{E}\left[\left(\sum_{i=1}^{q}\alpha_i Y_{n-i}+W_n\right)W_n\right]\\ &= \sum_{i=1}^{q}\alpha_i \underbrace{\mathcal{E}[Y_{n-i}W_n]}_{0}+R_W(0)=R_W(0)\end{aligned}$$

$$\therefore\quad R_Y(0) = \sum_{i=1}^{q}\alpha_i R_Y(i)+R_W(0)$$

$$\begin{aligned}R_Y(k) &= \mathcal{E}\left[Y_{n-k}\left(\sum_{i=1}^{q}\alpha_i Y_{n-i}+W_n\right)\right]\\ &= \sum_{i=1}^{q}\alpha_i R_Y(k-i)+\underbrace{\mathcal{E}[Y_{n-k}W_n]}_{\underbrace{\mathcal{E}[Y_{n-k}]\mathcal{E}[W_n]}_{0}}\\ &= \sum_{i=1}^{q}\alpha_i R_Y(k-i)\end{aligned}$$

b)

$$\begin{aligned}Y_n &= rY_{n-1}+W_n\\ R_Y(0) &= rR_Y(1)+R_W(0)\\ R_Y(k) &= rR_Y(k-1)\Rightarrow R_Y(1)=rR_Y(0)\\ \Rightarrow R_Y(0) &= r^2R_Y(0)+R_W(0)\Rightarrow R_Y(0)=\frac{R_W(0)}{1-r^2}\\ \Rightarrow R_Y(k) &= \begin{cases}\frac{r^kR_W(0)}{1-r^2}=\underbrace{\left(\frac{R_W(0)}{1-r^2}\right)}_{\sigma_Y^2}r^k & k>0\\ R_Y(-k)=\sigma_Y^2r^{|k|} & k<0\end{cases}\end{aligned}$$

10.3 Bandlimited Random Processes

10.45)

a) Suppose you have a signal $x(t)$, which is bandlimited and has a fourier transform $X(f)$, $|f|<W$. if you sample it with rate T_s you have

$$s(t) = \sum_{n=-\infty}^{+\infty} \delta(t-nT_s)$$

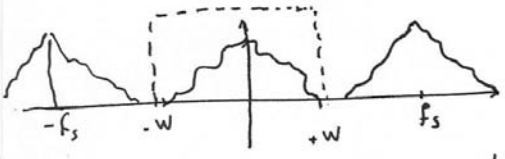
$$\hat{x}(t) = s(t)x(t) = x(t)\sum_{-\infty}^{+\infty}\delta(t-nT_s)$$

$$\hat{X}(f) = X(f) * \mathcal{F}\left\{\sum_{-\infty}^{+\infty}\delta(t-nT_s)\right\}$$

it can be easily shown that fourier of a δ train is a δ train, since δ train in t, is a periodic signal. so we have:

$$\hat{X}(f) = X(f) * \left[\frac{1}{T_s}\sum_{n=-\infty}^{+\infty}\delta(f-nf_s)\right] = \frac{1}{T_s}\sum_{n=-\infty}^{+\infty}X(f-nf_s)$$

Since $x(f)$ is bandlimited to $|f|<W$, $X(f)$ can be recovered from $\hat{x}(f)$ if we use an Ideal LP filter with bandwidth W, and if $f_s > 2W$



This sampling rate is called Nyquist rate.

$$s(t) = \sum_{n=-\infty}^{+\infty}\delta(t-nT_s) = \frac{1}{T_s}\sum_{k=-\infty}^{+\infty}e^{jk\omega_0 T} \quad \text{(Fourier series)}$$

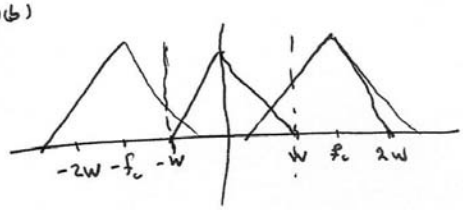
Therefore:

$$S(f) = \frac{1}{T_s}\sum_{n=-\infty}^{+\infty}\delta(f-nf_s)$$

P10.45)

b) Assume $W < f_s < 2W$

Therefore, from figure 10.10(b)



Therefore the error part would be:

$$e(f) = X(f-f_s)\left[1-u(f-W)\right] + X(f+f_s)\left[u(f+W)\right]$$

and:

$$\hat{X}(f)_{\text{recovered}} = f_s\left[X(f)+e(f)\right]$$

for $0 < f_s < 2W$, $e(f) = \left[\sum_{k=1}^{\infty} X(f-kf_s)\right]\left[1-u(f-W)\right] + u(f+W)\sum_{k=1}^{\infty} X(f+f_s)$

P10.45)

c) $\hat{X}(t) = X(t)\,s(t)$

$$E\left[\hat{X}(t)\hat{X}(t+\tau)\right] = E\left[X(t)X(t+\tau)\,s(t)\,s(t+\tau)\right] = s(t)\,s(t+\tau)\,R_X(\tau)$$

$$s(t) = \sum_{n=-\infty}^{+\infty} \delta(t-nT_s)$$

$$s(t+\tau) = \sum_{m=-\infty}^{+\infty} \delta(t+\tau-nT_s)$$

$$s(t) * s(t+\tau) = \sum_{k=-\infty}^{+\infty} \delta(\tau-kT_s)$$

Therefore: $R_{\hat{X}}(\tau) = R_X(\tau)\sum_{k=-\infty}^{+\infty} \delta(\tau-kT_s)$

And: $S_{\hat{X}}(f) = S_X(f) * f_s\sum_{n=-\infty}^{+\infty} \delta(f-nf_s) = f_s\sum_{n=-\infty}^{+\infty} S_X(f-nf_s)$

P10.45)

d) just like part b:

Assume $W < f_s < 2W$

Therefore $S_{X_{recovered}}(f) = f_s \left[S_X(f) + e(f) \right]$

in which $e(f) = S_X(f - f_s)\left[1 - u(f - W)\right] + S_X(f + f_s)\left[u(f + W)\right]$

for any $f_s < 2W$: $e(f) = \left[\sum_{k=1}^{\infty} S_X(f - kf_s) \right]\left[1 - u(f - W)\right] + \sum_{k=1}^{\infty} S_X(f + kf_s)\, u(f + W)$

e)

$$S_X(f) = \begin{cases} A + \frac{A}{W} f, & -W < f < 0 \\ A - \frac{A}{W} f, & 0 \le f < W \end{cases}$$

if $W < f_s < 2W$

$$\Rightarrow e(f) = S_X(f - f_s)\left[1 - u(f - W)\right] + S_X(f + f_s)\left[u(f + W)\right]$$

$$\int_{-\infty}^{+\infty} e(f)\, df = \int_{-W}^{-f_s + W} \left(A - \frac{A}{W}(f + f_s)\right) df + \int_{f_s - W}^{W} \left(A + \frac{A}{W}(f - f_s)\right) df$$

$$= \frac{1}{2} \times \frac{W}{A}\left(A - \frac{A}{W}(f_s - W)\right)^2 + \frac{1}{2} \times \frac{W}{A}\left(A + \frac{A}{W}(W - f_s)\right)^2 = \frac{W}{A}\left(2A - A\frac{f_s^2}{W}\right) = WA\left(2 - \frac{f_s}{W}\right)$$

10.46) $H(f) = \begin{cases} 1 & |f| < f_c < 1/2 \\ 0 & f_c < |f| < 1/2 \end{cases}$

a)

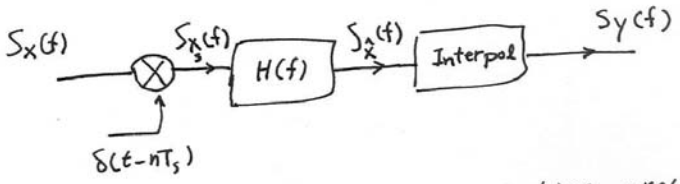
inverse fourier transform:

$$h(n) = \int_{-f_c}^{f_c} e^{j2\pi fn}\, df = \int_{-f_c}^{f_c} (\cos 2\pi fn + j\sin 2\pi fn)\, df$$

$$= \frac{2}{2\pi n} \sin 2\pi f_c n = \frac{\sin 2\pi f_c n}{\pi n}$$

b) Approach:

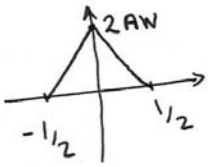
$S_Y(f)$



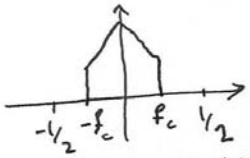
After sampling with Nyquist rate, the discrete time process has periodic a shape, with base period like the original signal

Therefore:

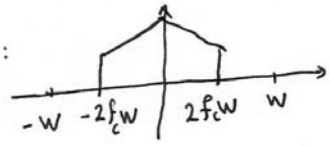
$S_{X_s}(f)$:



Therefore $S_{\hat{X}}(f) = |H(f)|^2 S_{X_s}(f)$:



and interpolator achieves the equivalent continuous time process with following $S_Y(f)$:



P10.46)

b) Approach 2: , let $X_s(n) = X(nT_s)$, $R_{X_s}(k) = R_X(kT_s)$

$$R_{\hat{X}}(k) = h(n) * h(-n) * R_{X_s}(k)$$

$$= \sum_{m=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} h(m)h(l) R_{X_s}(k-l+m)$$

$$R_Y(\tau) = \sum_{k=-\infty}^{+\infty} R_{\hat{X}}(k) p(t-kT_s) = \sum_{m=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} h(m)h(l) \sum_{k=-\infty}^{+\infty} R_{X_s}(k-l+m)p(t-kT_s)$$

Therefore $R_Y(\tau) = \sum_{-\infty}^{+\infty} \sum_{-\infty}^{+\infty} h(m)h(l) R_X(\tau-(l-m)T_s)$

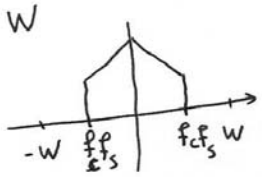
So, we have

$$S_Y(f) = \sum_{m=-\infty}^{+\infty} h(m) e^{-j2\pi T_s m f} \sum_{l=-\infty}^{+\infty} h(l) e^{j2\pi T_s l f} S_X(f)$$

$$= |H(fT_s)|^2 S_X(f) = \left|H\left(\frac{f}{f_s}\right)\right|^2 S_X(f) \quad , f_s = \frac{1}{2W}$$

if $H\left(\frac{f}{f_s}\right) = \begin{cases} 1 & |f| < |f_c| f_s < W \\ 0 & f_c f_s < |f| < W \end{cases}$ $\quad f_s = \frac{1}{2W}$

Then: $S_Y(f)$ would be:



10.47 1)

a) $H(f) = \frac{j2\pi f}{T} \quad |f| < \frac{1}{2}$

$$h(n) = \int_{-1/2}^{1/2} H(f)\, e^{j2\pi fn}\, df = \frac{1}{T}\int_{-1/2}^{1/2} j2\pi f\, e^{j2\pi fn}\, df$$

$$= \frac{j2\pi}{T}\left[\frac{1}{j2\pi n} f e^{j2\pi fn} - \frac{1}{j2\pi n} \times \frac{1}{j2\pi n} e^{j2\pi fn}\right]_{-\frac{1}{2}}^{\frac{1}{2}}, \quad n \neq 0$$

$$= \frac{j2\pi}{T}\left[\frac{1}{j4\pi n}\left(e^{j\pi n} + e^{-j\pi n}\right) + \frac{1}{(2\pi n)^2}\left(e^{j\pi n} - e^{-j\pi n}\right)\right]$$

$$= \frac{1}{T}\left[\frac{\cos \pi n}{n} - \frac{\sin \pi n}{\pi n^2}\right] = \frac{\pi n \cos \pi n - \sin \pi n}{T\pi n^2} = \frac{(-1)^n}{nT}, \quad n \neq 0$$

and $h(n) = 0, \; n = 0$

P10.47)

b)

$$x(t) = a\cos(2\pi f_0 t + \theta),\ \frac{1}{T} = 4f_0 \Rightarrow x(n) = x(nT) = a\cos\left(\frac{\pi}{2}n + \theta\right)$$

Therefore: $x_0(n) = x(n) * h(n)$ or $X_0(f) = X(f)H(f)$

$$X(f) = \frac{a}{2}\delta\left(f - \frac{1}{4}\right)\left[\cos\theta + j\sin\theta\right] + \frac{a}{2}\delta\left(f + \frac{1}{4}\right)\left[\cos\theta - j\sin\theta\right]$$

$$= \frac{a}{2}\left[\delta\left(f - \frac{1}{4}\right)e^{j\theta} + \delta\left(f + \frac{1}{4}\right)e^{-j\theta}\right]$$

$$H(f) = j\frac{2\pi f}{T}$$

Therefore:

$$X_0(f) = X(f)H(f) = \frac{j2\pi f}{T} \cdot \frac{a}{2}\left[\delta\left(f - \frac{1}{4}\right)e^{j\theta} + \delta\left(f + \frac{1}{4}\right)e^{-j\theta}\right]$$

$$= \frac{j\pi a}{4T}\left[\delta\left(f - \frac{1}{4}\right)e^{j\theta} - \delta\left(f + \frac{1}{4}\right)e^{-j\theta}\right]$$

Therefore $$x_0(n) = \frac{j\pi a}{4T}\left[e^{j\frac{\pi}{2}n}e^{j\theta} - e^{-j\frac{\pi}{2}n - j\theta}\right]$$

$$= \frac{j\pi a}{4T}\left[+2j\sin\left(\frac{\pi}{2}n + \theta\right)\right] = \frac{-\pi a}{2T}\sin\left(\frac{\pi}{2}n + \theta\right)$$

$$X_0(nT) = x_0(n) = \frac{-\pi a}{2T}\sin(2\pi n f_0 T + \theta)$$

after interpolation $$Y(t) = \sum_{n=-\infty}^{+\infty}\frac{\pi a}{2T}\sin(2\pi f_0 nT + \theta)p(t - nT)$$

since $f_0 = \frac{1}{4T} < \frac{1}{T}$, then $Y(t) = \frac{-\pi a}{2T}\sin(2\pi f_0 T + \theta)$

10.48

we have

$$R_X(\tau) = \sum R_X(nT)\, p(\tau - nT)$$

define $R'_X(\tau) = R_X(\tau - \alpha)$, $R'_X(\tau)$ has the same spectrum as $R_X(\tau)$

So: $R'_X(\tau) = \sum R'_X(nT)\, p(\tau - nT)$

or

$$R_X(\tau - \alpha) = \sum R_X(nT - \alpha)\, p(\tau - nT) \qquad (1)$$

first we define $y(t) = E\left[(X(t_0) - \hat{X}(t_0))X(t)\right]$, where t_0 is a fixed tim

$$y(mT) = E\left[(X(t_0) - \hat{X}(t_0))X(mT)\right] = R_X(t_0 - mT) - \sum_n R_X(mT - nT)\, p(t_0 - nT)$$

From (1): $\sum_n R_X(mT - nT)\, p(t_0 - nT) = R_X(t_0 - mT)$

Therefore $y(mT) = 0$

now for general t we have:

$$y(t) = R_X(t_0 - t) - \sum_n R_X(t - nT)\, p(t_0 - nT)$$

Since $BW < \frac{1}{T} \Longrightarrow y(t) = R_X(t_0 - t) - R_X(t_0 - t_0) = 0$

Therefore $y(t) = 0$, $\forall t$

Now, if you let $t = t_0$, $y(t_0) = E\left\{(X(t_0) - \hat{X}(t_0))X(t_0)\right\} = 0$, $\forall t_0$

therefore $E\left[(X(t_0) - \hat{X}(t_0))X(t_0)\right] = 0$, $\forall t_0$

P10. 48) — continued —

Also: $E\left[(X(t)-\hat{X}(t))\hat{X}(t)\right]=2(t)$

we showed that $E[X(t)\hat{X}(t)]=E[X(t)X(t)]=R_X(0)$

Now we show that $E[\hat{X}(t)\hat{X}(t)]=R_X(0)$

To do so:

$$E\{\hat{X}(t)\hat{X}(t)\}=E\left\{\sum_n\sum_m X(t-nT)X(nT)p(t-nT)p(t-mT)\right]$$

$$=\sum_n\sum_m R_X(mT-nT)\,p(t-nT)p(t-mT)$$

using ①

$$=\sum_n R_X(t-nT)p(t-nT)=R_X(0)$$

Therefore $E[(X(t)-\hat{X}(t))\hat{X}(t)]=R_X(0)-R_X(0)=0$

And $\hat{X}(t)=X(t)$ in Mean Square Sense.

10.49

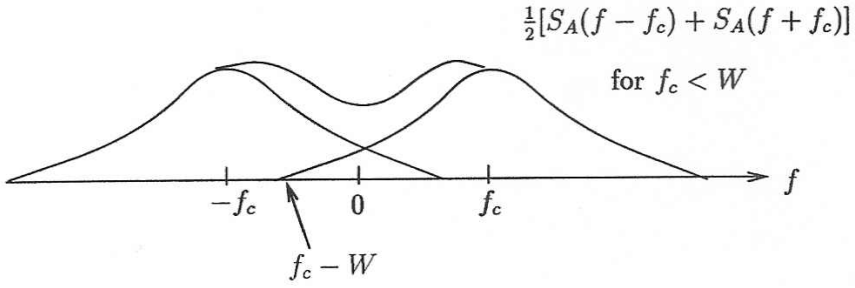
~~7.37~~ $Y(t) = \underbrace{A(t)\cos(2\pi f_c t + \Theta)}_{X(t)} + N(t)$

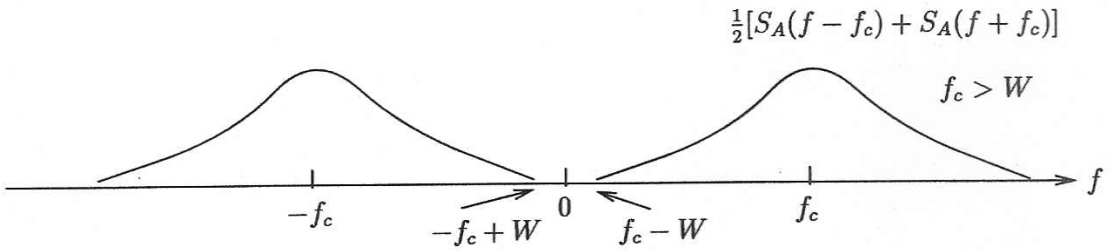
Assuming $X(t)$ and $N(t)$ are independent random processes:

$$R_Y(\tau) = R_X(\tau) + R_N(\tau) ,$$

from Example10.4 and the fact that $\mathcal{E}[X(t)] = 0$.

$$\begin{aligned} S_Y(f) &= S_X(f) + S_N(f) \\ &= \frac{1}{2}S_A(f - f_c) + \frac{1}{2}S_A(f + f_c) + S_N(f) \end{aligned}$$



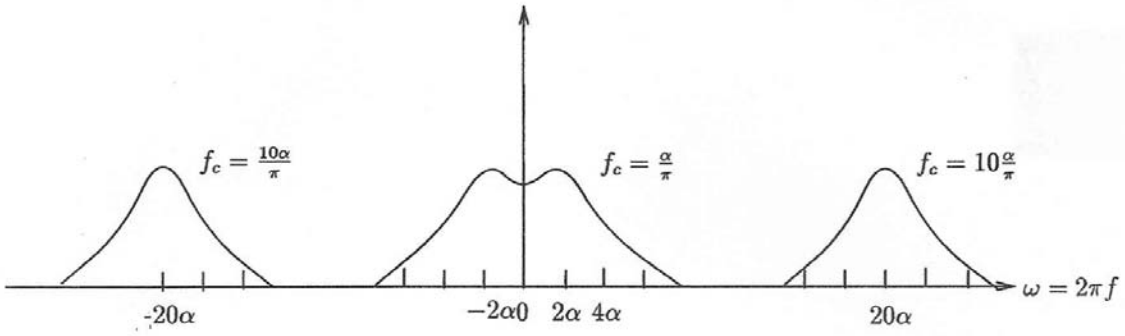


where we assumed that $S_A(f)$ is bandlimited to $|f| < W$.

10.50

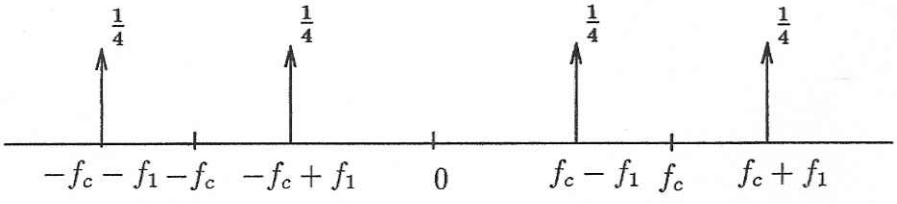
7.38 For the random telegraph signal:

$$\begin{aligned} S_X(f) &= \frac{4\alpha}{4\alpha^2 + 4\pi^2 f^2} \\ S_Y(f) &= \frac{1}{2}S_X(f+f_c) + \frac{1}{2}S_X(f-f_c) \\ &= \frac{2\alpha}{4\alpha^2 + 4\pi^2(f+f_c)^2} + \frac{2\alpha}{4\alpha^2 + 4\pi^2(f-f_c)^2} \\ &= \frac{2\alpha}{4\alpha^2(\omega + 2\pi f_c)^2} + \frac{2\alpha}{4\alpha^2 + (\omega - 2\pi f_c)^2} \end{aligned}$$

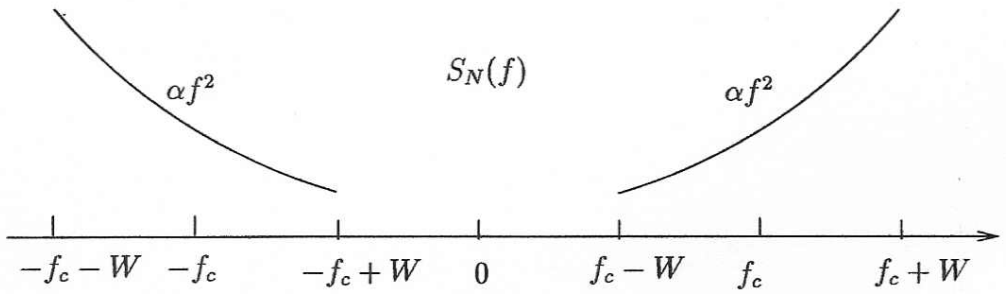


10.51

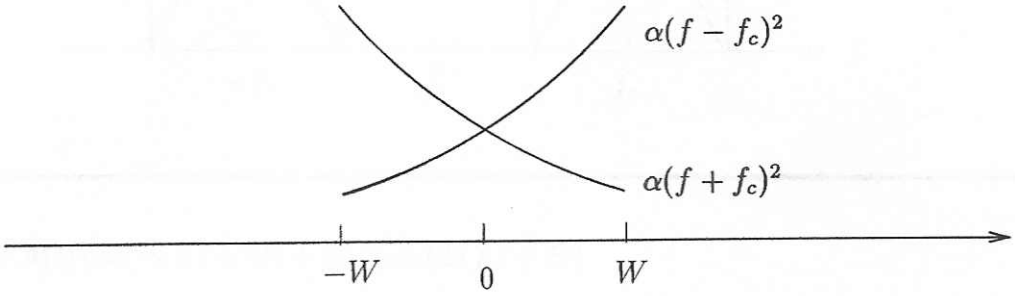
$$\begin{aligned} A(t) &= 2\cos(2\pi f_1 + \Phi) \\ R_A(\tau) &= \cos 2\pi f_1 \tau \qquad S_A(f) = \frac{1}{2}\delta(f+f_1) + \frac{1}{2}\delta(f-f_1) \\ S_X(f) &= \frac{1}{2}S_A(f+f_c) + \frac{1}{2}S_A(f-f_c) \\ &= \frac{1}{4}[\delta(f+f_c+f_1) + \delta(f+f_c-f_1) \\ &\quad +\delta(f-f_c+f_1) + \delta(f-f_c-f_1)] \end{aligned}$$



10.52
~~7.40~~



$$S_{N_c}(f) = [S_N(f - f_c) + S_N(f + f_c)]_2 = \alpha(f - f_c)^2 + \alpha(f + f_c)^2 \qquad |f| < W$$

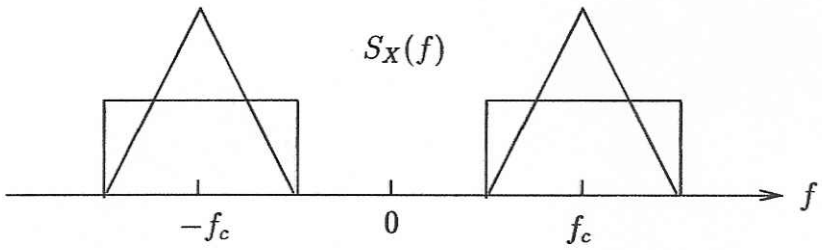


$$\begin{aligned}
\sigma_{N_c}^2 &= \int_{-W}^{W} \alpha(f - f_c)^2 df + \int_{-W}^{W} \alpha(f + f_c)^2 df \\
&= \frac{2}{3}[(f_c + W)^3 - (f_c - W)^3] \\
&= \frac{2}{3}W^3 + r f_c^2 W \\
\text{SNR} &= \frac{\sigma_X^2}{4f_c^2 W + \frac{2}{3}W^3}
\end{aligned}$$

10.53

$$X(t) = A(t)\cos(2\pi f_c t + \Theta) + B(t)\sin(2\pi f_c t + \Theta)$$

$$\begin{aligned}
\mathcal{E}[X(t)X(t+\tau)] &= \mathcal{E}[A(t)A(t+\tau)\cos(2\pi f_c t + \Theta)\cos(2\pi f_c t + 2\pi f_c \tau + \Theta)] \\
&\quad + \mathcal{E}[A(t)B(t+\tau)\cos(2\pi f_c t + \Theta)\sin(2\pi f_c t + 2\pi f_c \tau + \Theta)] \\
&\quad + \mathcal{E}[A(t+\tau)B(t)\cos(2\pi f_c t + 2\pi f_c \tau + \Theta)\sin(2\pi f_c t + \Theta)] \\
&\quad + \mathcal{E}[B(t)B(t+\tau)\sin(2\pi f_c t + \Theta)\sin(2\pi f_c t + 2\pi f_c \tau + \Theta)] \\
&= \frac{1}{2}R_A(\tau)\cos 2\pi f_c \tau + \frac{1}{2}R_B(\tau)\cos 2\pi f_c \tau \\
&= \frac{1}{2}(R_A(\tau) + R_B(\tau))\cos 2\pi f_c \tau \\
S_X(f) &= \frac{1}{4}[S_A(f+f_c) + S_B(f+f_c) + S_A(f-f_c) + S_A(f+f_c)]
\end{aligned}$$

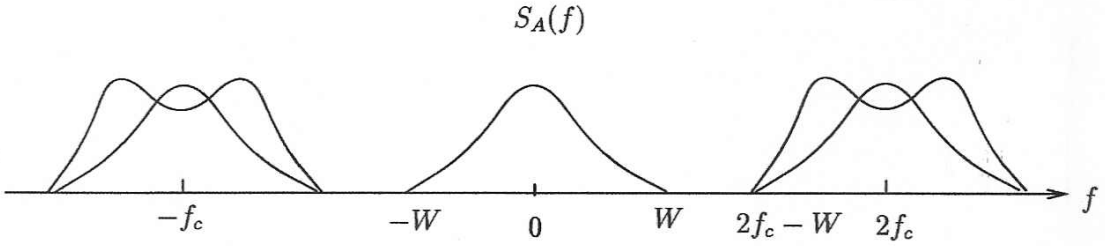


10.54

7.42 $X(t) = A(t)\cos(2\pi f_c t + \Theta) + B(t)\sin(2\pi f_c t + \Theta)$

$$\begin{aligned} 2X(t)\cos(\omega_c t + \Theta) &= 2A(t)\cos^2(\omega_c t + \Theta) \\ &\quad +2B(t)\cos(\omega_c t + \Theta)\sin(\omega_c t + \Theta) \\ &= 2A(t)\left(\frac{1}{2} + \frac{1}{2}\cos(2\omega_c t + 2\Theta)\right) \\ &\quad +2B(t)\frac{1}{2}\sin(2\omega_c t + 2\Theta) \\ &= A(t) + A(t)\cos(2\omega_c t + 2\Theta) \\ &\quad +B(t)\sin(2\omega_c t + 2\Theta) \end{aligned}$$

The power spectral density of this signal is shown below:



As long as $2f_c - W > W$, then the output of the ideal LPF is $A(t)$.
The same condition guarantees the recovery of $B(t)$.

10.55

$$\begin{aligned} R_{BA}(\tau) &= \mathcal{E}[B(t+\tau)A(t)] \\ &= \mathcal{E}[A(t)B(t+\tau)] \\ &= R_{AB}(-\tau) \end{aligned}$$

Eqn. 10.67b implies

$$R_{AB}(-\tau) = R_{BA}(\tau) = -R_{AB}(\tau)$$

$\Rightarrow R_{AB}(\tau)$ is an odd function of τ. Then

$$\begin{aligned} S_{AB}(f) &= \mathcal{F}[R_{AB}(\tau)] \\ &= \int_{-\infty}^{\infty} R_{AB}(\tau)e^{-j2\pi f\tau}d\tau \\ &= \underbrace{\int_{-\infty}^{\infty} R_{AB}(\tau)\cos 2\pi f\tau d\tau}_{0} - j\int_{-\infty}^{\infty} R_{AB}(\tau)\sin 2\pi f\tau d\tau \\ &= -j\int_{-\infty}^{\infty} R_{AB}(\tau)\sin 2\pi f\tau d\tau \end{aligned}$$

$\therefore S_{AB}(f)$ is a purely imaginary, odd form of f.
From Prob10.7 we also have

$$S_{BA}(f) = S_{AB}^{*}(f) = j\int_{-\infty}^{\infty} \alpha_{AB}(\tau)\sin 2\pi f\tau d\tau$$

10.4 Optimum Linear Systems

10.56

~~7.44~~ $X_\alpha = Z_\alpha + N_\alpha$ where $R_Z(k) = \sigma_Z^2 r^{|k|}$

$$\begin{bmatrix} 1+\Gamma & r \\ r & 1+\Gamma \end{bmatrix} \begin{bmatrix} h_0 \\ h_1 \end{bmatrix} = \begin{bmatrix} 1 \\ r \end{bmatrix}$$

where $\Gamma = \dfrac{\sigma_N^2}{\sigma_Z^2}$.

$$\begin{bmatrix} h_0 \\ h_1 \end{bmatrix} = \frac{1}{(1+\Gamma)^2 - r^2} \begin{bmatrix} 1+\Gamma & -r \\ -r & 1+\Gamma \end{bmatrix} \begin{bmatrix} 1 \\ r \end{bmatrix} = \frac{1}{(1+\gamma)^2 - r^2} \begin{bmatrix} 1+\Gamma-r^2 \\ \Gamma r \end{bmatrix}$$

$$\begin{aligned} \mathcal{E}[(Z_t - Y_t)^2] &= R_Z(0) - \sum_{\beta=0}^{1} h_\beta R_{ZX}(\beta) = R_Z(0) - \sum_{\beta=0}^{1} h_\beta \sigma_Z^2 r^{|\beta|} \\ &= \sigma_Z^2 \left[1 - \frac{1-\Gamma-r^2}{(1+\Gamma)^2 - r^2} - \frac{\Gamma r}{(1+\Gamma)^2 - r^2} r \right] \\ &= \sigma_Z^2 \left[1 - \frac{(1+\Gamma)(1-r^2)}{(1+\Gamma)^2 - r^2} \right] = \frac{29}{64}\sigma_Z^2 \end{aligned}$$

since $\Gamma = \frac{1}{4}$ $r = \frac{3}{4}$.

10.57

~~7.45~~ $R_Z(k) = \sigma_Z^2 r_1^{|k|} \qquad R_N(k) = \sigma_N^2 r_2^{|k|}$

a) Eqn. 10.83 implies

$$\sigma_Z^2 r_1^{|m|} = \sum_{\beta=0}^{p} h_\beta \left\{ \sigma_Z^2 r_1^{|m-\beta|} + \sigma_N^2 r_2^{|m-\beta|} \right\}$$

b)
$$\begin{bmatrix} \sigma_Z^2+\sigma_N^2 & \sigma_Z^2 r_1+\sigma_N^2 r_2 & \cdots & \sigma_Z^2 r_1^p+\sigma_N^2 r_2^p \\ \sigma_Z^2 r_1+\sigma_N^2 r_2 & \sigma_Z^2+\sigma_N^2 & \cdots & \\ \vdots & & \ddots & \\ \sigma_Z^2 r_1+\sigma_N^2 r_2 & & \cdots & \sigma_Z^2+\sigma_N^2 \end{bmatrix} \begin{bmatrix} h_0 \\ h_1 \\ \vdots \\ h_p \end{bmatrix} = \sigma_Z^2 \begin{bmatrix} 1 \\ r_1 \\ r_1^2 \\ \vdots \\ r_1^p \end{bmatrix}$$

c) Let $p = 2$, $\underbrace{\sigma_Z^2 = 9 \quad r_1 = \frac{1}{3}}_{\text{decays slowly with } |k|}$ $\underbrace{\sigma_N^2 = 1 \quad r_2 = \frac{1}{3}}_{\text{decays quickly with } |k|}$

$$\begin{bmatrix} 10 & 9(\frac{1}{3})+1(\frac{1}{9}) & 9(\frac{1}{9})+1(\frac{1}{81}) \\ 6.3 & 10 & 6.3 \\ 4.11 & 6.3 & 10 \end{bmatrix} \begin{bmatrix} h_0 \\ h_1 \\ h_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ 4 \end{bmatrix}$$

$\Rightarrow h_0 = 0.867 \quad h_1 = 0.0383 \quad h_2 = 0.0189$

This is essentially a low pass filter.

d)
$$\begin{aligned} \mathcal{E}[e_t^2] &= R_Z(0) - \sum_{\beta=0}^{p} h_\beta R_Z(\beta) \\ &= 9 - 8.1165 \\ &= 0.8835 \end{aligned}$$

10.58

a) $R_Z(k) = \begin{cases} (1+\alpha^2)\sigma^2 \triangleq \sigma_Z^2 & k=0 \\ \alpha\sigma^2 & k=\pm 1 \\ 0 & ew \end{cases}$

Eqn. 10.83 is then $R_Z(m) = \sum_{\beta=0}^{p} h_\beta \left\{ R_Z(m-\beta) + \sigma_N^2 \delta_{m-\beta} \right\}$

$$\begin{bmatrix} \sigma_Z^2+\sigma_N^2 & \alpha\sigma^2 & 0 & 0 & \cdots & 0 \\ \alpha\sigma^2 & \sigma_Z^2+\sigma_N^2 & \alpha\sigma^2 & 0 & & \vdots \\ 0 & \alpha\sigma^2 & \sigma_Z^2+\sigma_N^2 & & & 0 \\ \vdots & & & & \ddots & \alpha\sigma^2 \\ 0 & & \cdots & 0 & \alpha\sigma^2 & \sigma_Z^2+\sigma_N^2 \end{bmatrix} \begin{bmatrix} h_0 \\ h_1 \\ \vdots \\ \\ h_p \end{bmatrix} = \begin{bmatrix} \sigma_Z^2 \\ \alpha\sigma^2 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

b) Let $p=2$ in previous matrix equation, and divide both sides by σ^2

$$\begin{bmatrix} 1+\alpha^2+\Gamma & & \\ \alpha & 1+\alpha^2+\Gamma & \alpha \\ 0 & \alpha & 1+\alpha^2+\Gamma \end{bmatrix} \begin{bmatrix} h_0 \\ h_1 \\ h_2 \end{bmatrix} = \begin{bmatrix} 1+\alpha^2 \\ \alpha \\ 0 \end{bmatrix}$$

where $\Gamma = \sigma_N^2/\sigma^2$. The last equation implies

$$h_2 = \frac{-\alpha}{1+\alpha^2+\Gamma} h_1$$

Then substituting for h_2 we obtain

$$\begin{bmatrix} 1+\alpha^2+\Gamma & \alpha \\ \alpha & \frac{(1+\alpha^2+\Gamma)^2-\alpha^2}{1+\alpha^2+\Gamma} \end{bmatrix} \begin{bmatrix} h_0 \\ h_1 \end{bmatrix} = \begin{bmatrix} 1+\alpha^2 \\ \alpha \end{bmatrix}$$

$$\begin{bmatrix} h_0 \\ h_1 \end{bmatrix} = \frac{1}{(1+\alpha^2+\Gamma)^2 - 2\alpha^2} \begin{bmatrix} \frac{(1-\alpha^2+\Gamma)^2-\alpha^2}{1+\alpha^2+\Gamma} & -\alpha \\ -\alpha & 1+\alpha^2+\Gamma \end{bmatrix} \begin{bmatrix} 1+\alpha^2 \\ \alpha \end{bmatrix}$$

$$h_0 = \frac{\frac{(1+\alpha^2+\Gamma)^2(1+\alpha^2)-\alpha^2(1+\alpha^2)}{1+\alpha^2+\Gamma} - \alpha^2}{(1+\alpha^2+\Gamma)^2 - 2\alpha^2}$$

$$h_1 = \frac{\alpha\Gamma}{(1+\alpha^2+\Gamma)^2 - 2\alpha^2}$$

$$h_2 = \frac{-\frac{\alpha^2\Gamma}{1+\alpha^2+\Gamma}}{(1+\alpha^2+\Gamma)^2 - 2\alpha^2}$$

c)

$$\begin{aligned} \mathcal{E}[e_t^2] &= R_Z(0) - \sum_{\beta=0}^{2} h_\beta R_Z(\beta) \\ &= R_Z(0) - h_0 R_Z(0) - h_1 R_Z(1) - \underbrace{h_2 R_Z(z)}_{0} \\ &= \sigma^2\{(1+\alpha^2)(1-h_0) - \alpha h_1\} \end{aligned}$$

Check: If $\Gamma = \sigma_N^2/\sigma^2 = 0$, i.e. no noise. Then $h_0 = 1$, $h_1 = 0$, $h_2 = 0$, i.e. no filtering and $\mathcal{E}[e_t^2] = 0$, i.e. no error.

10.59 $X_\alpha = Z_\alpha + N_\alpha$

a)
$$\begin{aligned} Y_t &= \sum_{\beta=-p}^{p} h_\beta X_{t-\beta} \\ R_{ZX}(m) &= \sum_{\beta=-p}^{p} h_\beta R_X(m-\beta) \qquad |m| \le p \\ R_{ZX}(m) &= \mathcal{E}[Z_n X_{n-m}] = \mathcal{E}[Z_n(Z_{n-m} + N_{n-m})] \\ &= R_Z(m) \end{aligned}$$

where we assume noise and desired signal are independent and noise is zero mean

$$R_X(m-\beta) = R_Z(m-\beta) + R_N(m-\beta)$$

$\therefore$ the optimum filter must satisfy

$$R_Z(m) = \sum_{\beta=-p}^{p} h_\beta \{R_Z(m-\beta) + R_N(m-\beta)\} \qquad |m| \le p$$

b)
$$\begin{bmatrix} R_Z(0)+R_N(0) & R_Z(1)+R_N(1) & \cdots & R_Z(2p)+R_N(2p) \\ R_Z(1)+R_N(1) & R_Z(0)+R_N(0) & \cdots & R_Z(2p-1)+R_N(2p-1) \\ \vdots & & & \vdots \\ R_Z(2p)+R_N(2p) & \cdots & & R_Z(0)+R_N(0) \end{bmatrix} \begin{bmatrix} h_{-p} \\ h_{-p+1} \\ \vdots \\ h_0 \\ \vdots \\ h_p \end{bmatrix}$$

$$= \begin{bmatrix} R_Z(-p) \\ \vdots \\ R_Z(0) \\ \vdots \\ R_Z(p) \end{bmatrix}$$

c)
$$\begin{bmatrix} 1+\Gamma & r & r^2 \\ r & 1+\Gamma & r \\ r^2 & r & 1+\Gamma \end{bmatrix} \begin{bmatrix} h_{-1} \\ h_0 \\ h_1 \end{bmatrix} = \begin{bmatrix} r \\ 1 \\ r \end{bmatrix}$$

$$r = \frac{3}{4}, \qquad \Gamma = \frac{\sigma_N^2}{\sigma^2} = \frac{1}{4}$$

$$\begin{bmatrix} \frac{5}{4} & \frac{3}{4} & \frac{9}{16} \\ \frac{3}{4} & \frac{5}{4} & \frac{3}{4} \\ \frac{9}{16} & \frac{3}{4} & \frac{5}{4} \end{bmatrix} \begin{bmatrix} h_{-1} \\ h_0 \\ h_1 \end{bmatrix} = \begin{bmatrix} \frac{3}{4} \\ 1 \\ \frac{3}{4} \end{bmatrix}$$

$$h_{-1} = .16438 \quad h_0 = .60274 \quad h_1 = .16438$$

d)
$$\begin{aligned} \mathcal{E}[e_t^2] &= R_Z(0) - \sum_{\beta=-1}^{1} h_\beta R_Z(\beta) \\ &= R_Z(0) - h_{-1}R_Z(-1) - h_0 R_Z(0) - h_1 R_Z(1) = 1.6028 \end{aligned}$$

10.60

7.48 a) $q\left(\frac{1}{3}\right)^m = q\sum_{\beta=1}^{p} h_\beta \left(\frac{1}{3}\right)^{|m-\beta|} \qquad m \in \{1,2,...,p\}$

b) $p = 2$:

$$\begin{bmatrix} 1 & \frac{1}{3} \\ \frac{1}{3} & 1 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} q \\ \frac{1}{3} q \end{bmatrix}$$

$$\Rightarrow h_1 = \frac{1}{3} \qquad h_2 = 0$$

c) $\mathcal{E}[e_g^2] = q - q\left(\frac{1}{3}\right) = \frac{2}{3}q$

10.61

7.49 $\hat{X}(t) = aX(t_1) + bX(t_2)$

a)
$$\begin{aligned} e(t) &= \hat{X}(t) - X(t) \\ &= aX(t_1) + bX(t_2) - X(t) \end{aligned}$$

Orthogonality condition implies that

$$\begin{aligned} \mathcal{E}[(aX(t_1) + bX(t_2) - X(t))X(t_1)] &= 0 \\ \mathcal{E}[(aX(t_1) + bX(t_2) - X(t))X(t_2)] &= 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow \quad aR_X(0) + bR_X(t_2 - t_1) &= R_X(t - t_1) \\ aR_X(t_1 - t_2) + bR_X(0) &= R_X(t - t_2) \end{aligned}$$

$$\Rightarrow \quad \begin{bmatrix} R_X(0) & R_X(t_2 - t_1) \\ R_X(t_2 - t_1) & R_X(0) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} R_X(t - t_1) \\ R_X(t - t_2) \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{R_X^2(0) - R_X^2(t_2 - t_1)} \begin{bmatrix} R_X(0)R_X(t - t_1) - R_X(t_2 - t_1)R_X(t - t_2) \\ R_X(0)R_X(t - t_2) - R_X(t_2 - t_1)R_X(t - t_1) \end{bmatrix}$$

b)
$$\begin{aligned} \mathcal{E}[e^2(t)] &= \mathcal{E}[e(t)[aX(t_1) + bX(t_2) - X(t)]] \\ &= \underbrace{a\mathcal{E}[e(t)X(t_1)]}_{0} + \underbrace{b\mathcal{E}[e(t)X(t_2)]}_{0} - \mathcal{E}[e(t)X(t)] \\ &= -a\mathcal{E}[X(t_1)X(t)] - b\mathcal{E}[X(t_2)X(t)] + \mathcal{E}[X(t)X(t)] \end{aligned}$$

$$\begin{aligned} \mathcal{E}[e^2(t)] = R_X(0) &- \frac{R_X(0)R_X(t - t_1) - R_X(t_2 - t_1)R_X(t - t_2)}{R_X^2(0) - R_X^2(t_2 - t_1)} R_X(t - t_1) \\ &- \frac{R_X(0)R_X(t - t_2) - R_X(t_2 - t_1)R_X(t - t_1)}{R_X^2(0) - R_X^2(t_2 - t_1)} R_X(t - t_2) \end{aligned}$$

Check: If $t = t_1$ then $a = 1$, $b = 0$ and $\mathcal{E}[e^2(t)] = 0$.

10.62

7.50 $t_1 = t - d \qquad t_2 = t + d \qquad t_2 - t_1 = 2d$

$$\begin{aligned} a &= \frac{R_X(0)R_X(d) - R_X(2d)R_X(d)}{R_X^2(0) - R_X^2(2d)} = \frac{R_X(0) - R_X(2d)}{R_X^2(0) - R_X^2(2d)} R_X(d) \\ &= \frac{R_X(d)}{R_X(0) + R_X(2d)} \\ b &= \frac{R_X(0)R_X(d) - R_X(2d)R_X(d)}{R_X^2(0) + R_X(2d)} = \frac{R_X(d)}{R_X(0) + R_X(2d)} = a \\ \mathcal{E}[e^2(t)] &= R_X(0) - a[R_X(d) + R_X(d)] \\ &= R_X(0) - \frac{2R_X^2(d)}{R_X(0) + R_X(2d)} \end{aligned}$$

10.63

7.51 $t_1 = t - d \qquad t_2 = t - 2d \qquad t_1 - t_2 = d \qquad t - t_1 = d \qquad t - t_2 = 2d$

$$\begin{aligned} a &= \frac{R_X(0)R_X(d) - R_X(d)R_X(2d)}{R_X^2(0) - R_X^2(d)} = \frac{e^{-\alpha d} - e^{-\alpha 3d}}{1 - e^{-\alpha 2d}} = e^{-\alpha d} \\ b &= \frac{R_X(0)R_X(2d) - R_X(d)R_X(d)}{R_X^2(0) - R_X^2(d)} = \frac{e^{-\alpha 2d} - e^{-\alpha 2d}}{1 - e^{-\alpha 2d}} = 0 \\ \mathcal{E}[e^2(t)] &= R_X(0) - aR_X(d) = 1 - e^{-\alpha 2d} \end{aligned}$$

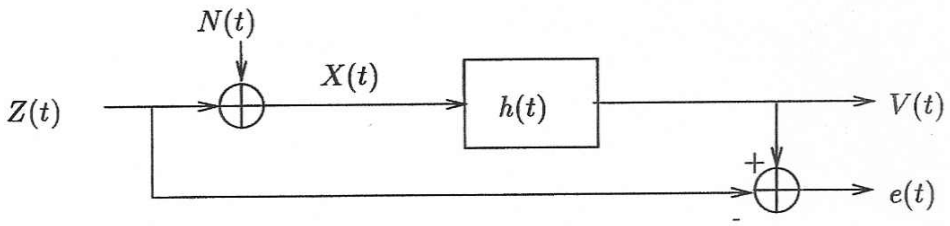
The optimum predictor of the form $\hat{X}(t) = aX(t-d)$ must satisfy

$$\begin{aligned} \mathcal{E}[e(t)X(t-d)] &= \mathcal{E}[(aX(t-d) - X(t))X(t-d)] = 0 \\ aR_X(0) &= R_X(d) \\ a &= e^{-\alpha d} \end{aligned}$$

This is the same predictor as obtained above.

Thus in this example (i.e. when $R_X(\tau) = e^{-\alpha|\tau|}$), the optimum predictor uses only the most recent observations.

10.64
~~7.52~~



In Problem10.30 we considered the above system. After making adjustments for the difference in notation, we have

$$S_e(f) = |1 - H(f)|^2 S_Z(f) + |H(f)|^2 S_N(f)$$

Equation10.92 implies that

$$\begin{aligned}
|1 - H(f)|^2 &= \left(\frac{S_N(f)}{S_Z(f) + S_N(f)}\right)^2 \\
\therefore \quad S_e(f) &= \frac{S_N^2(f)S_Z(f)}{(S_Z(f) + S_N(f))^2} + \frac{S_N(f)S_Z^2(f)}{(S_Z(f) + S_N(f))^2} \\
&= \frac{S_N(f)S_Z(f)}{S_Z(f) + S_N(f)} \\
\mathcal{E}[e^2(t)] &= R_e(0) = \int_{-\infty}^{\infty} \frac{S_N(f)S_Z(f)}{S_Z(f) + S_N(f)} df.
\end{aligned}$$

10.65

~~7.53~~ If $Z(t)$ is the random telegraph signal and $N(t)$ is white noise, then

$$\begin{aligned}
S_Z(f) &= \frac{4\alpha}{4\alpha^2 + 4\pi^2 f^2} \qquad S_N(f) = \frac{N_0}{2} \\
H(f) &= \frac{S_Z(f)}{S_Z(f) + S_N(f)} = \frac{\frac{4\alpha}{4\alpha^2+4\pi^2 f^2}}{\frac{4\alpha}{4\alpha^2+4\pi^2 f^2} + \frac{N_0}{2}} \\
&= \frac{4\alpha}{4\alpha + \frac{N_0}{2}(4\alpha^2 + 4\pi^2 f^2)} \\
&= \frac{4\alpha}{(4\alpha + 2N_0\alpha^2) + \frac{N_0}{2}4\pi^2 f^2}
\end{aligned}$$

10.64
From Problem ~~7.52~~

$$\begin{aligned}
\mathcal{E}[e^2(t)] &= \int_{-\infty}^{\infty} \frac{\frac{N_0}{2}4\alpha}{(4\alpha + 2N_0\alpha^2) + \frac{N_0}{2}4\pi^2 f^2} df \qquad x = \sqrt{2N_0}\pi f \\
&= \frac{2N_0\alpha}{\sqrt{2N_0}\pi} \int_{-\infty}^{\infty} \frac{1}{(4\alpha + 2N_0\alpha^2) + x^2} dx \qquad a^2 = 4\alpha + 2N_0\alpha^2 \\
&= \frac{\sqrt{2N_0}\alpha}{\pi\sqrt{4\alpha + 2N_0\alpha^2}} \underbrace{\int_{-\infty}^{\infty} \frac{a}{a^2 + x^2} dx}_{\pi} = \sqrt{\frac{2N_0\alpha}{4 + 2N_0\alpha}}
\end{aligned}$$

10.66

~~7.54~~ $S_Z(f) = \frac{N_1}{2} \qquad |f| < W \qquad S_N(f) = \frac{N_0}{2}$

$$H(f) = \frac{S_Z(f)}{S_Z(f) + S_N(f)} = \frac{\frac{N_1}{2}}{\frac{N_1}{2} + \frac{N_0}{2}} = \frac{N_1}{N_1 + N_0} \qquad |f| < w$$

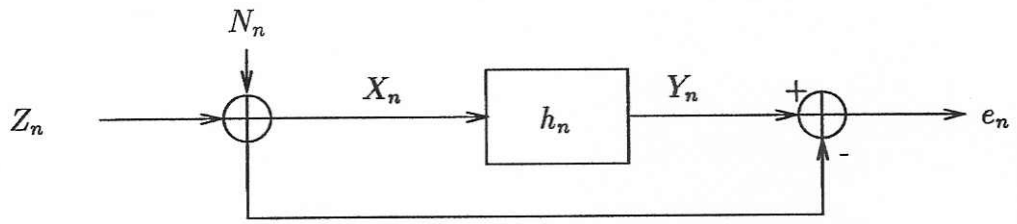
$$\mathcal{E}[e^2(t)] = \int_{-\infty}^{\infty} \frac{S_Z(f)S_N(f)}{S_Z(f) + S_N(f)} df = \frac{\frac{N_1 N_2}{2}}{N_1 + N_0} 2W = \frac{N_1 N_0}{N_1 + N_0} W$$

10.67

~~7.55~~ $S_Z(f) = \dfrac{2}{1 + 4\pi^2 f^2} \qquad S_N(f) = 1$

$$\begin{aligned} H(f) &= \frac{\frac{2}{1+4\pi^2 f^2}}{\frac{2}{1+4\pi^2 f^2} + 1} = \frac{2}{2 + 1 + 4\pi^2 f^2} = \frac{2}{3 + 4\pi^2 f^2} \\ \mathcal{E}[e^2(t)] &= \int_{-\infty}^{\infty} \frac{S_Z(f)S_N(f)}{S_Z(f) + S_N(f)} df = \int_{-\infty}^{\infty} \frac{2}{3 + 4\pi^2 f^2} df \qquad x = 2\pi f \\ &= \frac{1}{\pi} \frac{1}{\sqrt{3}} \int_{-\infty}^{\infty} \frac{\sqrt{3}}{3 + x^2} dx = \frac{1}{\sqrt{3}} = 0.577 \end{aligned}$$

10.68



Equation 10.89 states that the optimum filter is given by

$$\begin{aligned} H(f) &= \frac{S_{ZX}(f)}{S_X(f)} \\ &= \frac{S_Z(f)}{S_Z(f)+S_N(f)} \\ R_{ZX}(k) &= \mathcal{E}[Z_{n+k}(Z_n+X_n)] \\ &= R_Z(k) \\ \Rightarrow S_{ZX}(f) &= S_Z(f) \end{aligned}$$

From Example 10.14

$$\begin{aligned} S_Z(f) &= \frac{\sigma^2}{(1-\alpha e^{-j2\pi f})(1-\alpha e^{j2\pi f})} \\ S_Z(f)+S_N(f) &= \frac{\sigma^2}{(1-\alpha e^{-j2\pi f})(1-\alpha e^{j2\pi f})} + \sigma_N^2 \\ &= \frac{\sigma^2+\sigma_N^2(1-\alpha e^{-j2\pi f})(1-\alpha e^{j2\pi f})}{(1-\alpha e^{-j2\pi f})(1-\alpha e^{j2\pi f})} \\ \Rightarrow H(f) &= \frac{\sigma^2}{\sigma^2+\sigma_N^2(1-\alpha e^{j2\pi f})(1-\alpha e^{j2\pi f})} \\ \mathcal{E}[e_n^2] &= R_e(0) \end{aligned}$$

Proceeding as in Problem 10.64 we can show that

$$\begin{aligned} S_e(f) &= \frac{S_N(f)S_Z(f)}{S_N(f)+S_Z(f)} = S_N(f)H(f) \\ &= \frac{\sigma^2}{\frac{\sigma^2}{\sigma_N^2}+(1-\alpha e^{-j2\pi f})(1-\alpha e^{j2\pi f})} \end{aligned}$$

We need to find $R_e(k) = \mathcal{F}^{-1}[S_e(f)]$

Consider the denominator of $S_e(f)$; let $Z = e^{j2\pi f}$ and $\Gamma = \dfrac{\frac{\sigma^2}{\sigma_N^2}+1+\alpha^2}{\alpha}$

$$\frac{\sigma^2}{\sigma_N^2}+1+\alpha^2-\alpha e^{-j2\pi f}-\alpha e^{j2\pi f} = -\alpha Z^{-1}[Z^2-\Gamma Z+1]$$

$$
\begin{aligned}
&= -\alpha Z^{-1}[Z - Z_1][Z - Z_2] \text{ where } Z_i = \frac{\Gamma \pm \sqrt{\Gamma^2 - 4}}{2} \\
&= -\alpha[1 - Z_1 Z^{-1}[Z - Z_2] \\
&= \alpha Z_2[1 - Z_1 Z^{-1}][1 - \frac{1}{Z_2} Z] \\
&= \alpha Z_2[1 - Z_1 Z^{-1}][1 - Z_1 Z] \text{ since } Z_1 = \frac{1}{Z_2} \\
&= \alpha Z_2[1 - Z_1 e^{-j2\pi f}][1 - Z_1 e^{j2\pi f}] \\
\therefore S_e(f) &= \frac{\sigma^2}{\alpha Z_2[1 - Z_1 e^{-j2\pi f}][1 - Z_1 e^{j2\pi f}]} \\
\therefore R_e(k) &= \mathcal{F}^{-1}[S_e(f)] \\
&= \frac{\sigma^2 Z_1}{\alpha(1 - Z_1^2)} \mathcal{F}^{-1}\left[\frac{1 - Z_1^2}{(1 - Z_1 e^{-j2\pi f})(1 - Z_1 e^{j2\pi f})}\right] \\
&= \frac{\sigma^2 Z_1}{\alpha(1 - Z_1^2)} Z_1^{|k|}
\end{aligned}
$$

Finally we find that

$$\mathcal{E}[e_n^2] = R_e(0) = \frac{\sigma^2 Z_1}{\alpha(1 - Z_1^2)}$$

10.69

~~7.57~~ $S_Z(f) = \dfrac{4\alpha}{4\alpha^2 + 4\pi^2 f^2}$

$$\begin{aligned} S_X(f) &= S_Z(f) + S_N(f) \\ &= \frac{4\alpha}{4\alpha^2 + 4\pi^2 f^2} + \frac{N_0}{2} = \frac{N_0}{2}\frac{4\pi^2 f^2 + 4\alpha^2 + \frac{8\alpha}{N_0}}{4\pi^2 f^2 + 4\alpha^2} \\ &= \frac{N_0}{2}\frac{(j2\pi f + 2\beta)(-j2\pi f + 2\beta)}{(j2\pi f + 2\alpha)(j2\pi f + 2\alpha)} \qquad \beta = \alpha\sqrt{1 + \frac{2}{\alpha N_0}} \end{aligned}$$

Let

$$G(f) = \sqrt{N_0}2\frac{j2\pi f + 2\beta}{j2\pi f + 2\alpha}$$

Equation 10.99 is then

$$\begin{aligned} S_{ZX'}(f) &= \frac{S_{ZX}(f)}{G^*(f)} = \frac{4\alpha}{4\alpha^2 + 4\pi^2 f^2}\frac{2\alpha - j2\pi f}{2\beta - j2\pi f} \\ &= \frac{4\alpha}{(2\alpha + j2\pi f)(2\beta - j2\pi f)} \end{aligned}$$

$$= \frac{2\alpha}{\alpha + \beta}\left[\frac{1}{2\alpha + j2\pi f} + \frac{1}{2\beta - j2\pi f}\right]$$

The inverse transform is

$$R_{ZX'}(\tau) = \begin{cases} \frac{2\alpha}{\alpha+\beta}e^{-2\alpha\tau} & \tau > 0 \\ \frac{2\alpha}{\alpha+\beta}e^{2\beta\tau} & \tau < 0 \end{cases}$$

$$\therefore \quad H_2(f) = \mathcal{F}\left[\frac{2\alpha}{\alpha + \beta}e^{-2\alpha\tau}\mu(\tau)\right] = \frac{2\alpha/(\alpha + \beta)}{2\alpha + j2\pi f}$$

and

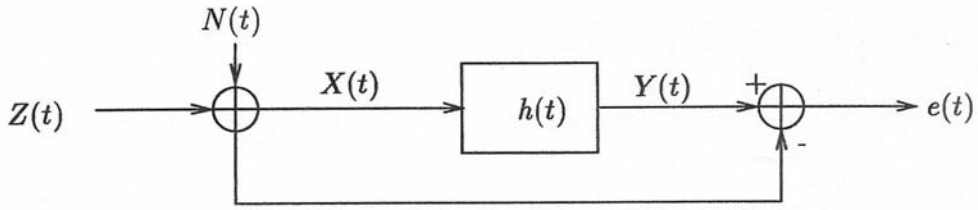
$$H(f) = \frac{H_2(f)}{G(f)} = \sqrt{\frac{2}{N_0}}\frac{2\alpha}{\alpha + \beta}\frac{1}{j2\pi f + 2\beta}$$

$$h(t) = \sqrt{\frac{2}{N_0}}\frac{2\alpha}{\alpha + \beta}e^{-\beta t} \qquad t \geq 0$$

where

$$\beta = \alpha\sqrt{1 + \frac{2}{\alpha N_0}}$$

10.70



In Problem 10.30 we considered the above system. After making adjustments for notation, we have

$$S_e(f) = |1 - H(f)|^2 S_Z(f) + |H(f)|^2 \underbrace{S_N(f)}_{1}$$

The Wiener filter in Example 10.25 is

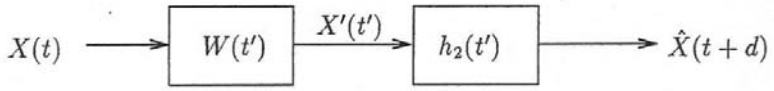
$$\begin{aligned}
H(f) &= \frac{c}{\sqrt{3} + j2\pi f} \text{ where } c = \frac{2}{1+\sqrt{3}} \\
1 - H(f) &= \frac{\sqrt{3} + j\omega - \frac{2}{1+\sqrt{3}}}{\sqrt{3} + j\omega} = \frac{1 + j\omega}{\sqrt{3} + j\omega} \\
S_e(f) &= \left|\frac{1 + j\omega}{\sqrt{3} + j\omega}\right|^2 \left|\frac{\sqrt{2}}{1 + j\omega}\right|^2 + \left|\frac{c}{\sqrt{3} + j\omega}\right|^2 \\
&= \frac{2 + c^2}{|\sqrt{3} + j\omega|^2} \\
R_e(\tau) &= \frac{1}{2\sqrt{3}}(2 + c^2)e^{-\sqrt{3}|t|} \\
R_e(0) &= \frac{2 + c^2}{2\sqrt{3}} = .732
\end{aligned}$$

As expected this is a larger error than the smoothing filter which uses the entire observation of $Z(\alpha)$.

10.71 First we find the causal filter that whitens the observation process

$$\begin{aligned} S_X(f) &= \mathcal{F}[e^{-\alpha|\tau|}] = \frac{2\alpha}{\alpha^2 + 4\pi^2 f^2} = \frac{\sqrt{2\alpha}}{\alpha + j2\pi f}\frac{\sqrt{2\alpha}}{\alpha - j2\pi f} \\ \Rightarrow W(f) &= \frac{\alpha + j2\pi f}{\sqrt{2\alpha}} \end{aligned}$$

Next we seek the optimum estimate for $X(t+d)$ in terms of the whitened process $X'(t)$



The estimator we seek is

$$\hat{X}(t+d) = \int_0^\infty h_2(\lambda_1) X'(t-\lambda) d\lambda$$

The orthogonality condition requires that

$$\begin{aligned} &\mathcal{E}[(\hat{X}(t+d) - X(t+d))X'(t')] = 0 \quad \text{for } t' < t \\ &\Rightarrow \mathcal{E}[\hat{X}(t+d)X'(t')] = \mathcal{E}[X(t+d)X'(t')] \qquad (*) \end{aligned}$$

The right hand side of the above equation is:

$$\begin{aligned} \mathcal{E}[X(t+d)X'(t')] &= \mathcal{E}\left[X(t+d)\int_0^\infty \omega(\lambda)X(t'-\lambda)d\lambda\right] \\ &= \underbrace{\int_0^\infty \omega(\lambda)R_X(t'-t-d-\lambda)d\lambda}_{\text{convolution of } \omega(\lambda) \star R_X(\lambda)'} \end{aligned}$$

$$\begin{aligned} \mathcal{F}[\omega(\lambda) \star R_X(\lambda)] &= W(f)S_X(f) = \frac{\alpha + j2\pi f}{\sqrt{2\alpha}}\frac{2\alpha}{\alpha^2 + 4\pi^2 f^2} = \frac{\sqrt{2\alpha}}{\alpha - j2\pi f} \\ \Rightarrow \omega(\tau) \star R_X(\tau) &= \sqrt{2\alpha}e^{\alpha\tau} \quad \tau < 0 \\ \Rightarrow \int_0^\infty \omega(\lambda)R_X(t'-t-d-\lambda)d\lambda &= \sqrt{2\alpha}e^{\alpha(t'-t-d)} \quad t'-t-d < 0 \end{aligned}$$

Now consider the left hand side of (*)

$$\begin{aligned} \mathcal{E}\left[\int_0^\infty h_2(\lambda)X'(t-\lambda)d\lambda X'(t')\right] &= \int_0^\infty h_2(\lambda)\underbrace{R_{X'}(t-t'-\lambda)d\lambda}_{S_{X'}(t-t'-\lambda)} \quad t' < t \\ &= h_2(t'-t) \quad t' < t \end{aligned}$$

Let $t'' = t - t'$

$$\begin{aligned} \therefore h_2(t'') &= \sqrt{2}e^{-\alpha d}e^{-\alpha t''} \quad t'' > 0 \\ \Rightarrow H_2(f) &= \frac{\sqrt{2}e^{-\alpha d}}{\alpha + j2\pi f} \end{aligned}$$

Finally the optimum filter is given by

$$\begin{aligned} H(f) &= W(f)H_2(f) = e^{-\alpha d} \\ \Rightarrow h(t) &= e^{-\alpha d}\delta(t) \\ \Rightarrow \hat{X}(t+d) &= e^{-\alpha d}X(t) \end{aligned}$$

10.72

7.60 $X_n = Z_n + N_n \qquad R_Z(k) = 4\left(\frac{1}{2}\right)^{|k|} \qquad R_N(k) = \delta_k$

$$S_X(f) = S_Z(f) + S_N(f) = \frac{4}{\frac{5}{4} - \cos 2\pi f} + 1 = \frac{\frac{21}{4} - \cos 2\pi f}{\frac{5}{4} - \cos 2\pi f}$$

$$= \frac{\frac{Z_2}{2}(1 - Z_2 e^{-j2\pi f})(1 - Z_1 e^{+j2\pi f})}{(1 - \frac{1}{2}e^{-j2\pi f})(1 - \frac{1}{2}e^{j2\pi f})}$$

after factoring the numerator and denominator where

$$Z_1 = \frac{\frac{21}{4} - \sqrt{\left(\frac{21}{4}\right)^2 - 4}}{2} \simeq \frac{1}{5}$$

$$Z_2 = \frac{\frac{21}{4} + \sqrt{\left(\frac{21}{4}\right)^2 - 4}}{2} = \frac{1}{Z_1}$$

$$\Rightarrow G(f) = \sqrt{\frac{Z_2}{2}} \frac{1 - Z_1 e^{-j2\pi f}}{1 - \frac{1}{2}e^{-j2\pi f}} = \frac{1}{W(f)}$$

Next consider:

$$R_{ZX}(k) = \mathcal{E}[Z_{n+n}(Z_n + N_n)] = R_Z(k)$$
$$\Rightarrow S_{ZX}(f) = S_Z(f)$$

Thus

$$\begin{aligned} S_{ZX'}(f) &= W^*(f) S_{ZX}(f) \\ &= \sqrt{\frac{2}{Z_2}} \frac{1 - \frac{1}{2}e^{+j2\pi f}}{1 - Z_1 e^{+j2\pi f}} S_Z(f) \\ &= \sqrt{\frac{2}{Z_2}} \left(\frac{1}{1 - Z_1 e^{+j2\pi f}}\right) \left(\frac{4}{1 - \frac{1}{2}e^{-j2\pi f}}\right) \\ &= \frac{4\sqrt{\frac{2}{Z_2}}}{1 - \frac{1}{2}Z_1} \left[\frac{1}{1 - Z_1 e^{j2\pi f}} + \underbrace{\frac{1}{1 - \frac{1}{2}e^{-j2\pi f}}}_{\text{this yields the positive time component}}\right] \text{ after partial fraction expansion} \end{aligned}$$

$$\therefore H_2(f) = \frac{4\sqrt{\frac{2}{Z_2}}}{1 - \frac{1}{2}Z_1} \frac{1}{1 - \frac{1}{2}e^{-j2\pi f}}$$

and finally

$$\begin{aligned} H(f) &= W(f)H_2(f) = \sqrt{\frac{2}{Z_2}} \frac{1 - \frac{1}{2}e^{-j2\pi f}}{1 - Z_1 e^{-j2\pi f}} \left(\frac{4\sqrt{\frac{2}{Z_2}}}{1 - \frac{1}{2}Z_1}\right) \left(\frac{1}{1 - \frac{1}{2}e^{-j2\pi f}}\right) \\ &= \frac{\frac{8}{Z_2(1 - \frac{1}{2}Z_1)}}{1 - Z_1 e^{-j2\pi f}} \\ &= \left(\frac{8}{Z_2 - \frac{1}{2}}\right) \frac{1}{1 - Z_1 e^{-j2\pi f}} \\ h_n &= \left(\frac{8}{Z_2 - \frac{1}{2}}\right) Z_1^k \quad k \geq 0 \end{aligned}$$

is the impulse response of the optimum filter.

*10.5 The Kalman Filter

10.73

7.63 $Z_n = \alpha_{n-1}Z_{n-1} + W_{n-1}$

$$\begin{aligned} P[Z_n|Z_{n-1}, Z_{n-2}, \ldots] &= P[W_{n-1}|Z_{n-1}, Z_{n-1}, \ldots] \\ &= P[W_{n-1}] \\ &= P[Z_n|Z_{n-1}] \end{aligned}$$

So Z_n is Markovian.

$$\begin{aligned} X_n &= Z_n + N_n \\ &= \alpha_{n-1}Z_{n-1} + W_{n-1} + N_n \\ &= \alpha_{n-1}(X_{n-1} - N_{n-1}) + W_{n-1} + N_n \\ &= \alpha_{n-1}X_{n-1} - \alpha N_{n-1} + N_n + W_{n-1} \end{aligned}$$

Similarly, X_n is Markovian.

10.74

7.64 From the equation just above (10.119 ~~7.114~~), we have

$$k_n(a_n - k_n)E[\mathcal{E}_n^2] = k_n^2 E[N_n^2]$$

$$\begin{aligned} E[\mathcal{E}_{n+1}^2] &= (a_n - k_n)^2 E[\mathcal{E}_n^2] + E[W_n^2] + k_n^2 E[N_n^2] \\ &= (a_n - k_n)^2 E[\mathcal{E}_n^2] + E[W_n^2] + k_n(a_n - k_n)E[\mathcal{E}_n^2] \\ &= a_n(a_n - k_n)^2 E[\mathcal{E}_n^2] + E[W_n^2] \end{aligned}$$

10.75

7.65 Initialization: $Y_0 = 0$, $E[\mathcal{E}_0^2] = E[Z_0^2] = 0$

$$\begin{aligned} k_n &= \frac{aE[\mathcal{E}_n^2]}{E[\mathcal{E}_n^2] + E[N - n^2]} = \frac{aE[\mathcal{E}_n^2]}{E[\mathcal{E}_n^2] + 1} \\ Y_{n+1} &= a_n Y_n + k_n(X_n - Y_n) \\ E[\mathcal{E}_{n+1}^2] &= a_n(a_n - k_n)E[\mathcal{E}_n^2] + E[W_n^2] \\ &= a\frac{a}{1 + E[\mathcal{E}_n^2]}E[\mathcal{E}_n^2] + 0.36 \end{aligned}$$

We then have

$$e_\infty = a\frac{a}{1 + e_\infty}e_\infty + 0.36$$

For $a = 0.5$, $e_\infty = 0.44$, $k_\infty = 0.15$

$$Y_{n+1} = 0.8Y_n + 0.15(X_n - Y_n)$$

For $a = 2$, $e_\infty = 3.46$, $k_\infty = 1.55$

$$Y_{n+1} = 0.8Y_n + 1.55(X_n - Y_n)$$

10.76

7.66 Define the innovation as

$$I_n = X_n - bY_n$$

We have

$$\begin{aligned} \mathcal{E}_{n+1} &= (a_n - k_n b_n)\mathcal{E}_n + W_n - k_n N_n \\ E[\mathcal{E}_{n+1}^2] &= (a_n - k_n b_n)^2 E[\mathcal{E}_n^2] + E[W_n^2] + k_n^2 E[N_n^2] \\ k_n &= \frac{a_n b_n E[\mathcal{E}_n^2]}{b_n^2 E[\mathcal{E}_n^2] + E[N - n^2]} \end{aligned}$$

And

$$\begin{aligned} Y_{n+1} &= aY_n + k_n(X_n - bY_n) \\ E[\mathcal{E}_{n+1}^2] &= a_n(a_n - k_n b_n)E[\mathcal{E}_n^2] + E[W_n^2] \end{aligned}$$

*10.6 Estimating the Power Spectral Density

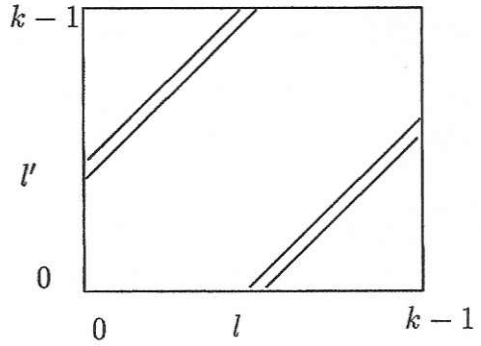
10.77

$$\begin{aligned}\tilde{p}_k(f) &= \frac{1}{k}\left|\sum_{l=0}^{k-1} X_l e^{-j2\pi fl}\right|^2 \\ &= \frac{1}{k}\sum_{l=0}^{k-1}\sum_{l'=0}^{k-1} X_l X_{l'} e^{-j2\pi fl} e^{+j2\pi fl'} \\ &= \frac{1}{k}\sum_{l=0}^{k-1}\sum_{l'=0}^{k-1} X_l X_{l'} e^{-j2\pi f(l-l')}\end{aligned}$$

$$m = l - l' \Leftrightarrow l' = l - m$$

where $(k-1) \leq m \leq k-1$

We change the order of summation as indicated by the figure below:



We then obtain:

$$= \sum_{m=-(k-1)}^{k-1} \left\{ \underbrace{\frac{1}{k}\sum_{l=0}^{k-|m|-1} X_l X_{l+n}}_{\hat{r}_k(m)} \right\} e^{-j2\pi fm}$$

$$= \sum_{m=-(k-1)}^{k-1} \hat{r}_k(m) e^{-j2\pi fm} \quad \surd$$

10.78

(a)
```
% Plots two 128-point periodograms
>x=fft(uniform_rnd(0,1,1,128));
>y=fft(uniform_rnd(0,1,1,128));
>plot((x.*conj(x))/128)
>hold on
>plot((y.*conj(y))/128)
```

(b)
```
x=fft(uniform_rnd(0,1,50,128));      50 periodograms
plot(mean(((x.*conj(x))/128)));      Averaged periodogram
```

10.79
```
% periodogram_AR.m
a=[1 -.5];
b=[1];
agg=zeros(1,128);
for i=1:50
 x=uniform_rnd(-1,1,1,128);
y=filter(b,a,x)
z=fft(y)
agg=agg+z.*conj(z)
end
plot(agg/50)
```

10.80

$$\mathcal{E}\left[\sum_{m=-(k-1)}^{k-1}\left(\frac{1}{k-|m|}\sum_{n=0}^{k-|m|-1}X_nX_{n+m}\right)e^{-j2\pi fm}\right]$$

$$=\sum_{m=(k-1)}^{k-1}\frac{1}{k-|m|}\left(\underbrace{\sum_{n=0}^{k-|m|-1}R_X(m)}_{(k-|m|)R_X(m)}\right)e^{-j2\pi fm}$$

$$=\sum_{m=-(k-1)}^{k-1}R_X(m)e^{-j2\pi fm}$$

The estimate is biased because the limits of the summation are finite.

10.7 Numerical Techniques for Processing Random Signals

10.81)

a) $f_s = 2kW_0$

$$e(f) = \left[\sum_{n=1}^{\infty} S_X(f - nf_s)\right](1 - u(f - kW_0)) + u(f + kW_0)\sum_{n=1}^{\infty} S_X(f + nf_s)$$

$$\int_{-\infty}^{+\infty} e(f) = \sum_{n=1}^{\infty} \int_{-kW_0}^{+kW_0} S_X(f + nf_s)\,df + \sum_{n=1}^{\infty} \int_{-kW_0}^{kW_0} S_X(f - nf_s)\,df$$

Since $S_X(f)$ is a like a Gaussian pdf function with mean 0 & variance W_0 we have

$$\int_{-\infty}^{+\infty} e(f)\,df = \beta^2 W_0 \sum_{n=1}^{\infty} \left(Q(nf_s - kW_0) - Q(nf_s + kW_0)\right)$$

$$+\beta^2 W_0 \sum_{n=1}^{\infty} \left(Q(nf_s - kW_0) - Q(nf_s + kW_0)\right) =$$

$$= 2\beta^2 W_0 \sum_{n=1}^{\infty} \left(Q(nf_s - kW_0) - Q(nf_s + kW_0)\right)$$

in order to make this error less than 1%, we neglect effects of $n > 1$, and therefore we have

$$Q(f_s - kW_0) - Q(f_s + kW_0) \approx Q(f_s - kW_0) = Q(2kW_0 - kW_0) = Q(kW_0) = 0.01$$

as a result $k > 2.3$

P 10.81)

b) $t = \frac{1}{Nf_0}$, $T = Mt_0 = \frac{1}{2f_0}$, $W = Mf_0 = \frac{1}{2T_0}$

if N is fixed, by increasing f_0, we decrease t_0, which means we are interested in high frequencies in the signal and therefore focusing on limited time interval of the signal.

c) $W = 2.3W_0$, $f_0 = \frac{W}{M} = \frac{2.3W_0}{\frac{N}{2}} = \frac{4.6W_0}{N}$, $t_0 = \frac{1}{4.6W_0}$ (independent of N)

$T = \frac{N}{9.2W}$

(d) The Octave code for part d:

```
%P10.81
%part d
clear all;
close all;
N=1024;
M=N/2;
W0=10;
t0=1/(4.6*W0);
f0=4.6*W0/N;
T=N/(9.2*W0);
m=-M:1:M-1;
Sx=exp(-1*(m.^2*f0^2)/(2*W0^2));
k=-M:1:M-1;
Rfex=fft(Sx);
Rf=Rfex.*conj(Rfex)/N;
Rex=Sx*exp(-j*2*pi*m'*k/N);
Rex2=f0*sqrt(Rex.*conj(Rex));
%Rx=sum(Sx*exp(-j*2*pi*m*k'/N));
Rx=exp(-1*(m.^2*t0^2)*(2*pi^2*W0^2))*(sqrt(2*pi)*W0);
subplot(2,1,1)
plot(m,Rx,m,Rex2);
legend('Actual','FFT');
subplot(2,1,2)
plot(Rx,Rex2);
title('Actual vs. FFT');
```

10.82

```
%P10.82
%Part a
clear all;
close all;
N=256;
M=N/2;
k=-M:1:M-1;
alpha=0.75;

%part a
Rx=4*alpha.^abs(k);
Sx=fft(Rx);
figure(1);
plot((k+M)/N,sqrt(Sx.*conj(Sx)));
title('P10.82a');


%part b
Rx=4*0.5.^abs(k)+16*0.25.^abs(k);
Sx=fft(Rx);
figure(2);
plot((k+M)/N,sqrt(Sx.*conj(Sx)));
title('P10.82b');


%part c
Rx=0.5*cos(2*pi*0.1*k);
Sx=fft(Rx);
figure(3);
plot((k+M)/N,sqrt(Sx.*conj(Sx)));
title('P10.82c');
```

10.83

```
%P10.83
%Part a
clear all;
close all;
N=256;
M=N/2;
k=-M:1:M-1;
fc=1/8;
Sx(1:1:N)= 1;
for i=1:1:N
    if (((i-M-1)/N) >= -fc)
        if (((i-M-1)/N) <= fc)
            Sx(i)= 0;
        end
    end
end
Rx=fft(Sx);
Rx2=sqrt(Rx.*conj(Rx))/N;
%note that Rx would be periodic with period N
figure(1);
plot(k+M,Rx2);
title('Problem 10.83a');


%Part b
Sx= 0.5+0.5*cos(2*pi*k/N);
Rx=fft(Sx);
Rx2=sqrt(Rx.*conj(Rx))/N;
%note that Rx would be periodic with period N
figure(2);
plot(k+M,Rx2);
title('Problem 10.83b');
```

10.84

```
%P10.84
%Part a,b,c
clear all;
close all;
N=256;
M=N/2;
k=-M:1:M-1;
fc=1/8;
alpha=0.25;
Rx=4*alpha.^abs(k)
Sx=fft(Rx);
Sx1=sqrt(Sx.*conj(Sx));
Sx2=Sx1;
fc = 1/4;
for i=1:1:N
    if (((i-M-1)/N) >= -fc)
        if (((i-M-1)/N) <= fc)
            Sx2(i) = 0;
        end
    end
end
figure(1);
plot((k+M)/N,Sx2);
title('Problem 10.84a');

%Part b
%Assume f0=0.1
Rx=0.5*cos(2*pi*0.1*k);
Sx=fft(Rx);
Sx1=sqrt(Sx.*conj(Sx));
Sx2=Sx1;
h2=-4*pi^2*(k/N).^2;
Sx2=Sx2.*h2;
figure(2);
plot((k+M)/N,Sx2);
title('Problem 10.84b');

%Part c
Rx=9*(1-abs(k)/3);
Rx(1:M-3)=0;
Rx(M+4:2*M)=0;
Sx=fft(Rx);
Sx1=sqrt(Sx.*conj(Sx));
figure(3)
plot((k+M)/N,Sx1);
title('Problem 10.84c');
```

10.85)

a)
$$R_X(\tau) = \int_{-\infty}^{+\infty} S_X(f) e^{-j2\pi\tau} df = \int_{-\infty}^{0} S_X(f) e^{-j2\pi f\tau} df + \int_{0}^{+\infty} S_X(f) e^{-j2\pi f} df$$

$$= \int_{0}^{+\infty} S_X(-f) e^{j2\pi f\tau} df + \int_{0}^{+\infty} S_X(f) e^{-j2\pi f\tau} df$$

$$= \int_{0}^{+\infty} \left[\left(S_X(f) e^{-j2\pi f\tau} \right)^* + \left(S_X(f) e^{-j2\pi f\tau} \right) \right] df$$

$$= 2\,\text{Real}\left\{ \int_{0}^{+\infty} S_X(f) e^{-j2\pi f\tau} df \right\}$$

we used the fact that if X is a real random process:

we have: $S_X(-f) = S_X^*(f)$

P10.85)

b)

$$R_X(\tau) = 2\mathrm{Re}\left\{ \int_0^{\infty} S_X(f) e^{-j2\pi f\tau} df \right\}$$

define $$y(\tau) := \int_0^{\infty} S_X(f) e^{-j2\pi f\tau} df \approx \int_0^{W} S_X(f) e^{-j2\pi f\tau} df$$

divide the range $(0,W)$ to N steps, therefore:

$$y(\tau) \approx \sum_{m=0}^{N-1} S_X(mf_0) e^{-j2\pi mf_0\tau} f_0$$

and dividing τ to N steps:

$$y(kt_0) \approx f_0 \sum_{m=0}^{N-1} S_X(mf_0) e^{-j2\pi mkf_0t_0}$$

in order to make the above equation an FFT, we should have:

$$t_0 f_0 = \frac{1}{N}.$$

Therefore $R_X(kt_0) = 2f_0\,\mathrm{Re}\{y(kt_0)\}$ $\quad 0 \le k < N-1$

also f_0 $\quad -N < k < 0$: $R_X(kt_0) = R_X(-kt_0)$, Assuming X is real

c) This approach gives a better resolution to $R_X(\tau)$ with the same amount of FFT computations.

10.86

a)

```
%P10.86
close all;
clear all;
n=0:1023;
X=normal_rnd(0,1,1,1024);
h=exp(-2*n);
Y=conv(X,h);
z=autocov(Y,30);
%theoretical autocovariance w
w=exp(-2*n)/(1-exp(-4));
plot(1:30,w(1:30),1:30,z(1:30));
title('Problem 10.86');
```

b) $R_X(k) = \begin{cases} \sigma_X^2 = 1 & k = 0 \\ 0 & k \neq 0 \end{cases}$

$R_Y(k) = R_X(k) * h(k) * h(-k) = \sigma_X^2 \cdot h(k) * h(-k)$

$h(k) = e^{-2k}, \; k \geq 0 \Rightarrow h(k) * h(-k) = \sum_{n=0}^{\infty} h(k) h(n+k) = e^{-2k} \sum_{n=0}^{\infty} e^{-4k} = \frac{e^{-2k}}{1-e^{-4}}$

$E[Y] = m_X(0) H(0) = 0$

Therefore $C_Y(k) = R_Y(k) = \frac{e^{-2k}}{1-e^{-4}}$

c) The output is also Gaussian, with the above auto covariance.

10.87

```
%p 10.87
close all;
clear all;
n=0:0.01:10.23;
r=exp(-2.*n);
%r=(-0.5).^n;
K=toeplitz(r);
[U,D,V]=svd(K);
X=normal_rnd(0,1,1,1024);
y=V*(D^0.5)*X';
plot(y);
z=autocov(y,200);
plot(1:200,r(1:200),1:200,z(1:200));
title('Problem 10.87');
```

Problems Requiring Cumulative Knowledge

10.88

7.71 In Example 9.38, $R_X(t_1,t_2) \neq R_X(t_1-t_2)$ so the process is not WSS. However, the process is WS cyclostationary so after stationarizing $X(t)$,

$$R_{X_S}(\tau) = \begin{cases} 1-\frac{|\tau|}{T} & |\tau| \leq T \\ 0 & |\tau| > T \end{cases}$$

and

$$S_X(f) = T\left(\frac{\sin \pi f T}{\pi f T}\right)^2$$

10.89

7.72 We perform Wiener filtering for the signal of Example10.26 to compare its operation and performance with Kalman filtering

$$\begin{aligned}
Z_n &= aZ_{n-1} + W_n, \quad a = 0.8, \quad \sigma_N^2 = 0.36 \\
S_Z(f) &= |H_Z(f)|^2\sigma_N^2 = \left|\frac{1}{1-ae^{-j2\pi f}}\right|^2 \sigma_N^2 = \frac{0.36}{1.64-1.6\cos 2\pi f} = \frac{0.225}{1.025-\cos 2\pi f} \\
R_Z(k) &= \frac{\sigma_N^2}{1-a^2}a^k = \frac{0.36}{1-0.64}a^k = a^k = (0.8)^k \\
X_n &= Z_n + N_n, \quad E[N_n^2] = 1 \\
S_X(f) &= S_Z(f) + S_N(f) = S_Z(f) + 1 = \frac{1.25-\cos 2\pi f}{1.025-\cos 2\pi f}
\end{aligned}$$

Since

$$c - \cos 2\pi f = \frac{1}{2b}(1-be^{-j2\pi f})(1-be^{j2\pi f}), \quad b = c \pm \sqrt{c^2-1}$$

then

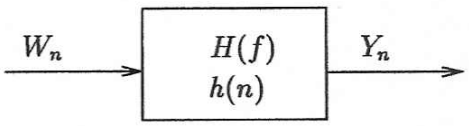
$$S_X(f) = \frac{(1-0.5e^{-j2\pi f})(1-0.5e^{j2\pi f})}{\frac{1}{1.6}(1-0.8e^{-j2\pi f})(1-0.8e^{j2\pi f})}$$

$$S_X(f) = G(f)\cdot G^*(f) \Rightarrow G(f) = \frac{1}{W(f)} = \sqrt{1.6}\frac{1-0.5e^{-j2\pi f}}{1-0.8e^{-j2\pi f}}$$

$$S_{ZX'}(f) = W^*(f)\cdot S_{ZX}(f), \quad S_{ZX}(f) = S_Z(f)$$

$$\begin{aligned}
\Rightarrow S_{ZX'}(f) &= \frac{1}{\sqrt{1.6}}\frac{1.08e^{j2\pi f}}{1-0.5e^{j2\pi f}} \cdot \frac{0.225}{\frac{1}{1.6}(1-0.8e^{-j2\pi f})(1-0.8e^{j2\pi f})} \\
S_{ZX'}(f) &= 0.225 \times \sqrt{1.6}\frac{1}{1-0.5e^{j2\pi f}} \cdot \frac{1}{1-0.8e^{-j2\pi f}} \\
&= 0.225 \times \sqrt{1.6} \times \frac{1}{1-0.5\times 0.8} \times \left[\frac{e^{j2\pi f}}{1-0.5e^{j2\pi f}} + \frac{1}{1-0.8e^{-j2\pi f}}\right] \\
\Rightarrow H_2(f) &= \frac{0.225\sqrt{1.6}}{1-0.4}\frac{1}{1-0.8e^{-j2\pi f}} \\
\Rightarrow H(f) &= W(f)\cdot H_2(f) = \frac{1}{\sqrt{1.6}}\frac{1-0.8e^{-j2\pi f}}{1-0.5e^{-j2\pi f}} \cdot \frac{0.225\sqrt{1.6}}{0.6}\frac{1}{1-0.8e^{-j2\pi f}} = \frac{0.375}{1-0.5e^{-j2\pi f}}
\end{aligned}$$

10.90 a) Let's first find the impulse response of the linear system



By taking the Fourier transform of both sides of the Y_n equation, we have

$$\begin{aligned}
\mathcal{F}[Y_n] &= -\sum_{i=1}^{q} \alpha_i \mathcal{F}[Y_{n-i}] + \sum_{i'=0}^{p} \beta_{i'} \mathcal{F}[W_{n-i'}] \\
Y(f) &= -\sum_{i=1}^{q} \alpha_i e^{-j2\pi i} Y(f) + \sum_{i'=0}^{p} \beta_{i'} e^{-j2\pi i'} W(f) \\
\Rightarrow H(f) &= \frac{Y(f)}{W(f)} = \frac{\sum_{i'=0}^{p} \beta_{i'} e^{-j2\pi i'}}{1 + \sum_{i=1}^{q} \alpha_i e^{-j2\pi i}}
\end{aligned}$$

For a linear system, we have $S_Y(f) = |H(f)|^2 \cdot S_W(f)$. Since W_n is white noise with variance σ_N^2, $S_W(f) = \sigma_W^2$

$$\Rightarrow S_Y(f) = \left| \frac{\Sigma \beta_{i'} e^{-j2\pi i'}}{1 + \Sigma \alpha_i e^{-j2\pi i}} \right|^2 \cdot \sigma_W^2$$

b) $Y_n W_{n-k} = -\Sigma \alpha_i Y_{n-i} W_{n-k} + \Sigma \beta_{i'} W_{n-i'} W_{n-k}$
Taking the expectations of both sides, we have:

$$\begin{aligned}
E[Y_n W_{n-k}] &= -\Sigma \alpha_i E[Y_{n-i} W_{n-k}] + \Sigma \beta_i E[W_{n-i'} W_{n-k}] \\
R_{YW}(-K) &= -\sum_i \alpha_i R_{YW}(i-k) + \sum_{i'} \beta_{i'} R_W(i'-k)
\end{aligned}$$

Let's take the Fourier transform of both sides with respect to k

$$\begin{aligned}
S_{YW}^*(f) &= -\sum_i \alpha_i e^{j2\pi i} S_{YW}^*(f) + \sum_{i'} \beta_{i'} e^{j2\pi i'} S_W^*(f) \\
\Rightarrow S_{YW}(f) &= \frac{\sum_{i'} \beta_{i'} e^{-j2\pi i'}}{1 + \Sigma \alpha_i e^{-j2\pi i}} S_W(f)
\end{aligned}$$

We assumed β_i and α_i's are real parameters

$$\begin{aligned}
Y_n Y_{n-k} &= -\Sigma \alpha_i Y_{n-i} Y_{n-k} + \Sigma \beta_{i'} W_{n-i'} Y_{n-k} \\
R_Y(-K) &= -\Sigma \alpha_i E[Y_{n-i} Y_{n-k} + \Sigma \beta_i R_{WY}(i'-k)] \\
S_Y^*(f) &= -\Sigma \alpha_i e^{j2\pi i} S_Y^*(f) + \Sigma \beta_{i'} e^{j2\pi i'} S_{WY}^*(f) \\
S_Y(f) &= \frac{\sum_{i'=0}^{p} \beta_{i'} e^{-j2\pi i'}}{1 + \sum_{i=1}^{q} \alpha_i e^{-j2\pi i}} S_{WY}(f), \quad S_{WY}(f) = S_{YW}^*(f) \\
\Rightarrow S_Y(f) &= \underbrace{\frac{\sum_{i'=0}^{p} \beta_{i'} e^{-j2\pi i'}}{1 + \sum_{i=1}^{q} \alpha_i e^{-j2\pi i}}}_{H(f)} \cdot \underbrace{\frac{\sum_{i'=0}^{p} \beta_{i'} e^{j2\pi i'}}{1 + \sum_{i=1}^{q} \alpha_i e^{j2\pi i}}}_{H^*(f)} S_W(f) \\
\Rightarrow S_Y(f) &= H(f) \cdot H^*(f) \cdot S_W(f) = |H(f)|^2 \cdot \sigma_W^2
\end{aligned}$$

10.91

7.75 a)
$$\begin{aligned} R_{Y_1Y_2}(\tau) &= E[Y_1(t+\tau)Y_2(t)] \\ &= E\left[\int_{-\infty}^{\infty} h_1(u)x_1(t+\tau-u)du \int_{-\infty}^{\infty} h_2(v)x_2(t-v)dv\right] \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_1(u)h_2(v)R_{X_1X_2}(\tau-u+v)dudv \\ &= h_1(\tau) \star h_2(-\tau)R_{X_1X_2}(\tau) \\ S_{Y_1Y_2}(f) &= H_1(f)H_2^*(f)S_{X_1X_2}(f) \end{aligned}$$

b) From Section 7.2, we know that $Y_1(t)$ and $Y_2(t)$ are WSS. From part (a), the cross-correlation only depends on $t_1 - t_2 = \tau$, so $Y_1(t)$ and $Y_2(t)$ are jointly WSS.

$Y_1(t)$ and $Y_2(t)$ are jointly Gaussian since $X_1(t)$ and $X_2(t)$ are jointly Gaussian, and the operation of convolution is linear.

c)
$$\begin{aligned} S_{Y_1Y_2}(f) &= H_1(f)H_2^*(f)S_{X_1X_2}(f) = 0 \\ R_{Y_1Y_2}(\tau) &= 0 \\ C_{Y_1Y_2}(\tau) &= 0 \end{aligned}$$

Y_1 and Y_2 are uncorrelated and therefore independent because they are jointly Gaussian.

d) $Y_1(t)$ and $Y_2(t)$ are still jointly Gaussian.

10.92)

a) $Y(t) = X(t) + N(t)$

$Y(t)$ should be sampled at points $t = mT$

$$X(t) = \sum_{K=-\infty}^{+\infty} a_K p(t - KT), \quad p(t) = \begin{cases} 1 & t = 0 \\ \emptyset & t = KT \end{cases}$$

Therefore the best point to look for a_K is at time KT.
and the sampling rate should be $\frac{1}{T}$

b) $Y(t) = X(t) + N(t) = \sum a_K p(t - KT) + N(t)$

$Y(KT) = a_K + N(KT)$, and $a_K = \begin{cases} 1 & \text{if data is 1} \\ 0 & \text{if data is 0} \end{cases}$

So, as can be seen, $N(KT)$ can change the sampled value from the actual value of a_K.
Therefore, we define a threshold as V, $(0 < V < 1)$ so that if $Y(KT) > V$, we assume the transmitted bit was 1, and we assume it was $\emptyset$, otherwise.

P10.92

c)

Probability of error:

$$P_{e_1} = P\{\text{we detect } \emptyset \text{ while bit was } 1\} = P\{Y_k < V \mid a_k = 1\} = P\{N_k < V - a_k \mid a_k = 1\}$$

$$= P\{N_k < V - 1 \mid a_k = 1\} = \int_{-\infty}^{V-1} P_N(y \mid a_{k=1})\, dy$$

in which P_N is pdf of noise $(N(t))$

$$P_{e0} = P\{\text{we detect } 1 \text{ while signal was } \emptyset\} = P\{Y_k > V \mid a_k = 0\} = \int_{V}^{\infty} P_N(y \mid a_{k=0})\, dy$$

Total error: $P_e = P_{e_1} \times P_{a_k=1} + P_{e0} \times P_{a_k=0}$

if N is independent of X and $P_{a_k=1} = P_{a_k=0} = \frac{1}{2}$

The best threshold would be $1/2$ and:

$$P_e = \frac{1}{2}\int_{-\infty}^{-1/2} P_N(y)\, dy + \frac{1}{2}\int_{\frac{1}{2}}^{\infty} P_N(y)\, dy = \frac{1}{2}\left[1 - \int_{-1/2}^{1/2} P_N(y)\, dy\right]$$